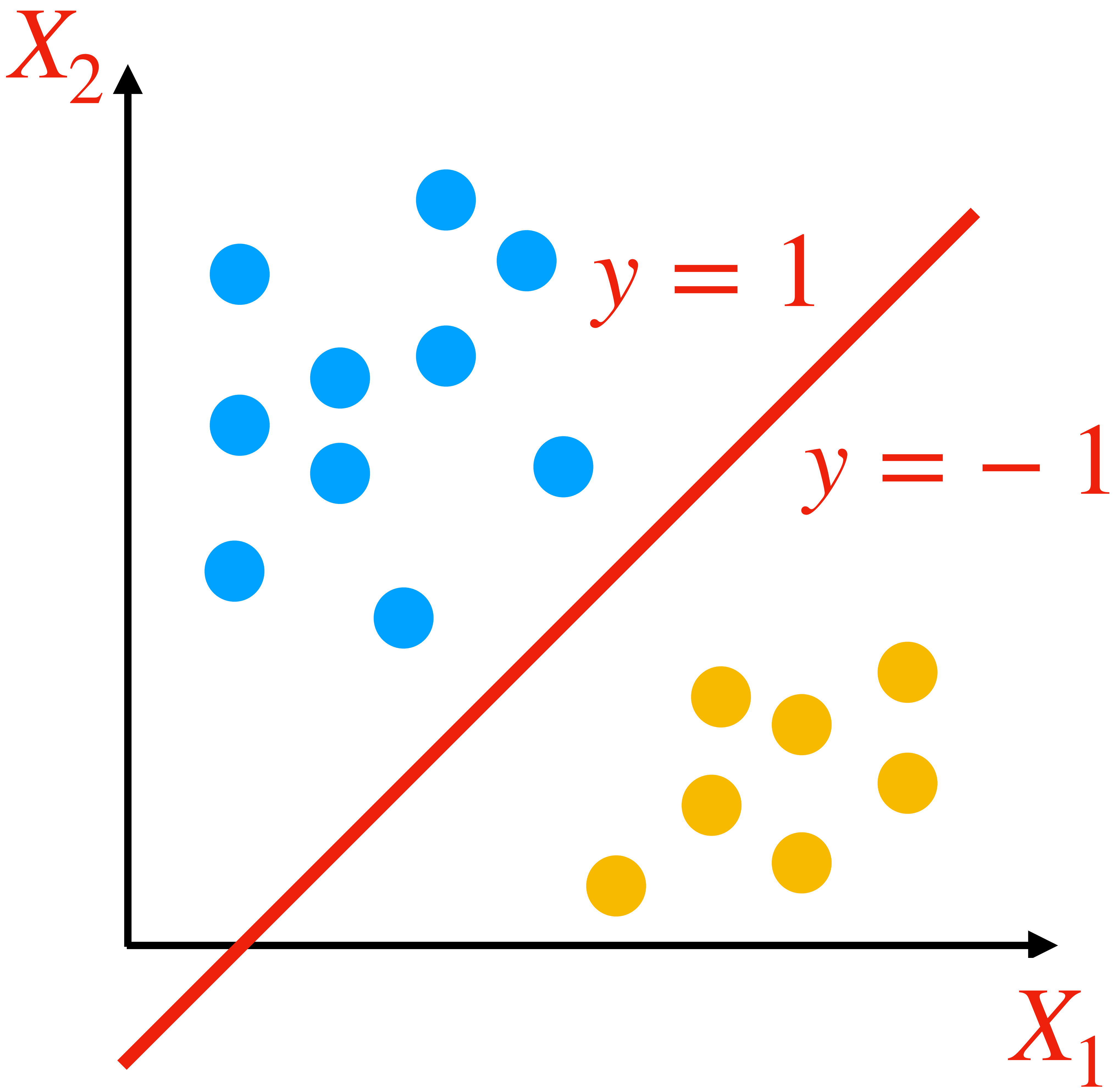
REVIEW OF PREVIOUS LECTURE: SUPPORT VECTOR MACHINES



We want to find the hyperplane that 'separates' the classes 'well'.

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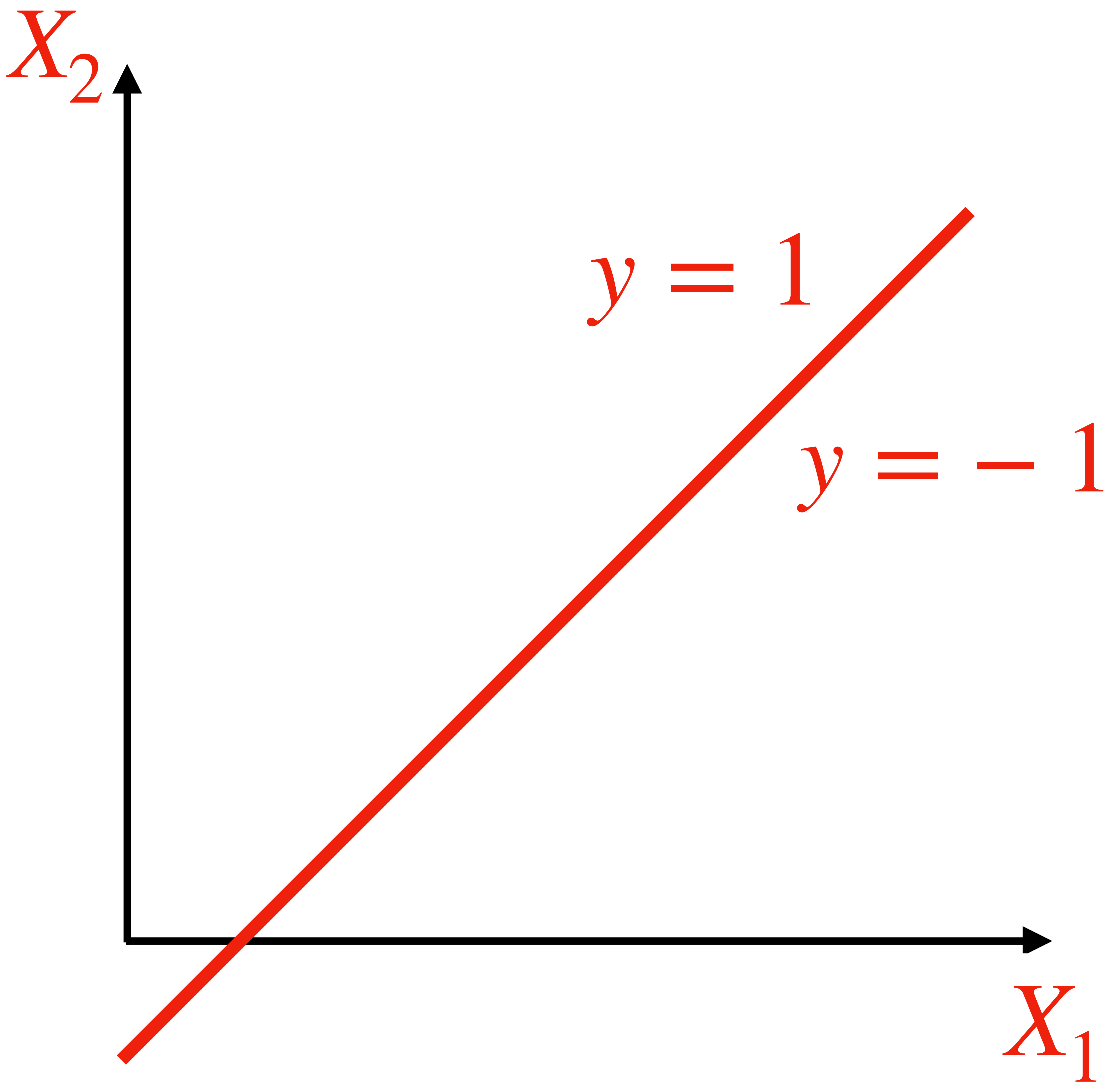

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Meaning of ‘separates classes’:

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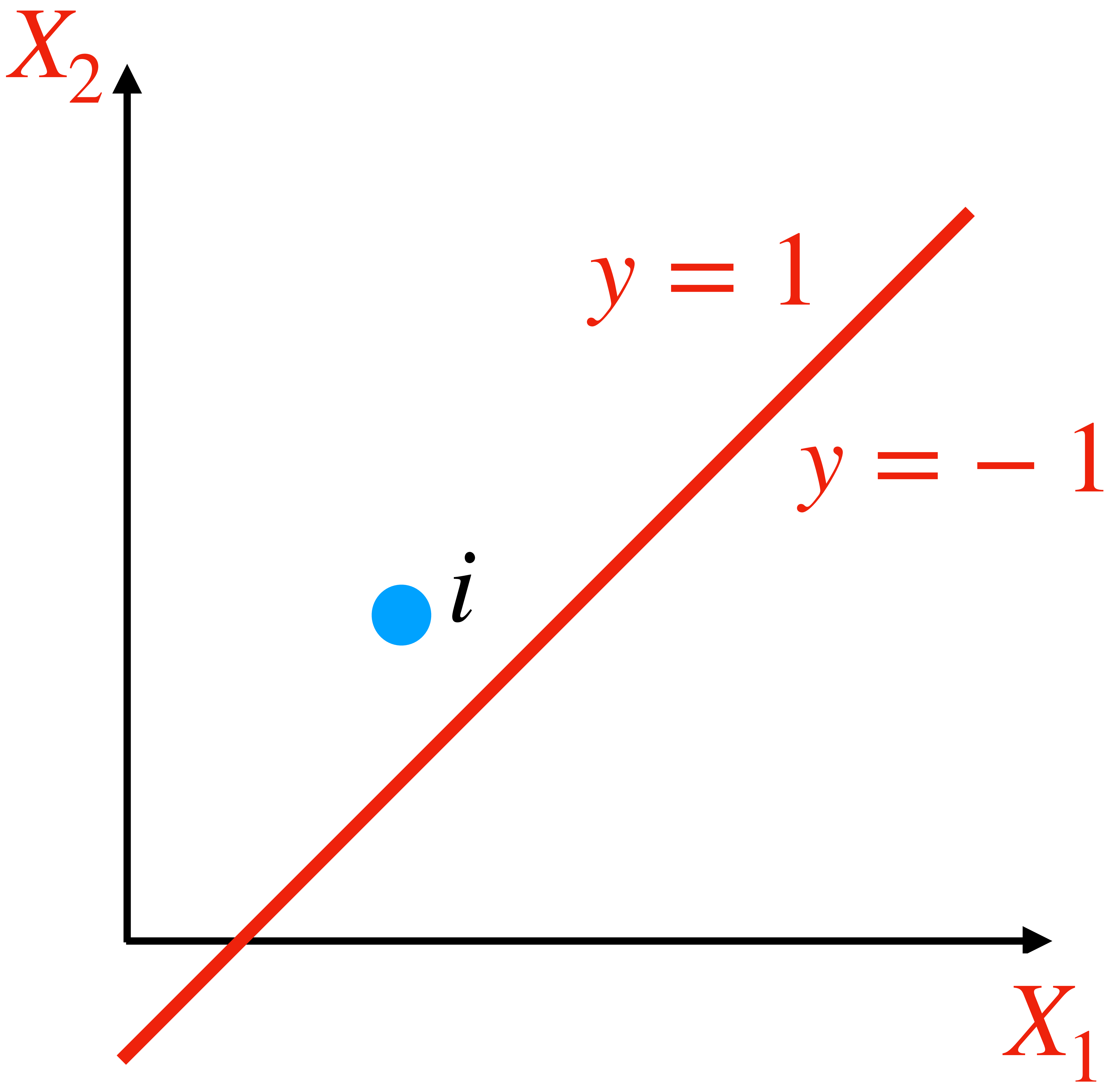

We want to find the hyperplane that 'separates' the classes 'well'.

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For a training point i with $y_i = 1$:

$$w_0 + w_1 x_{i1} + w_2 x_{i2} > 0$$

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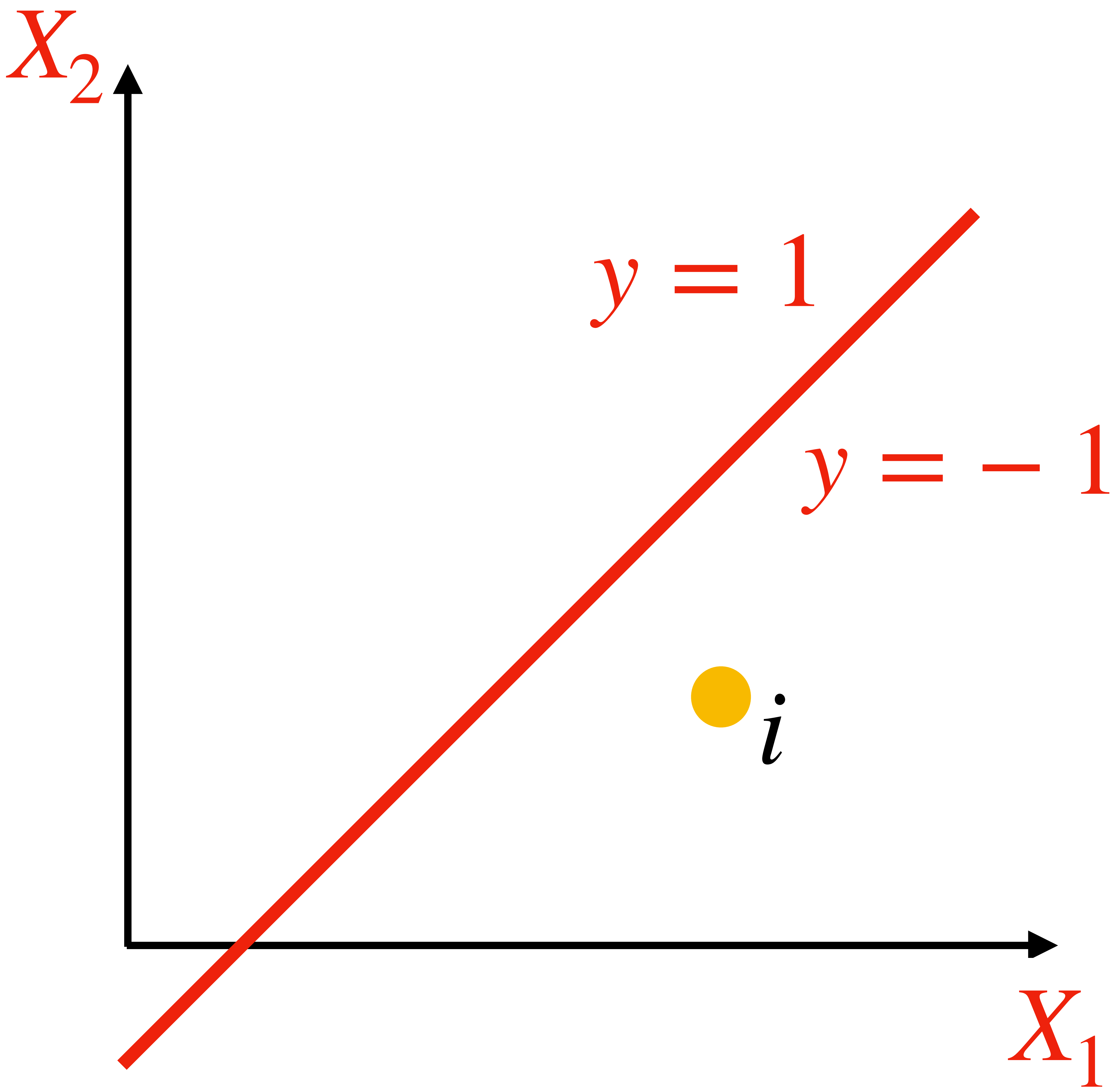

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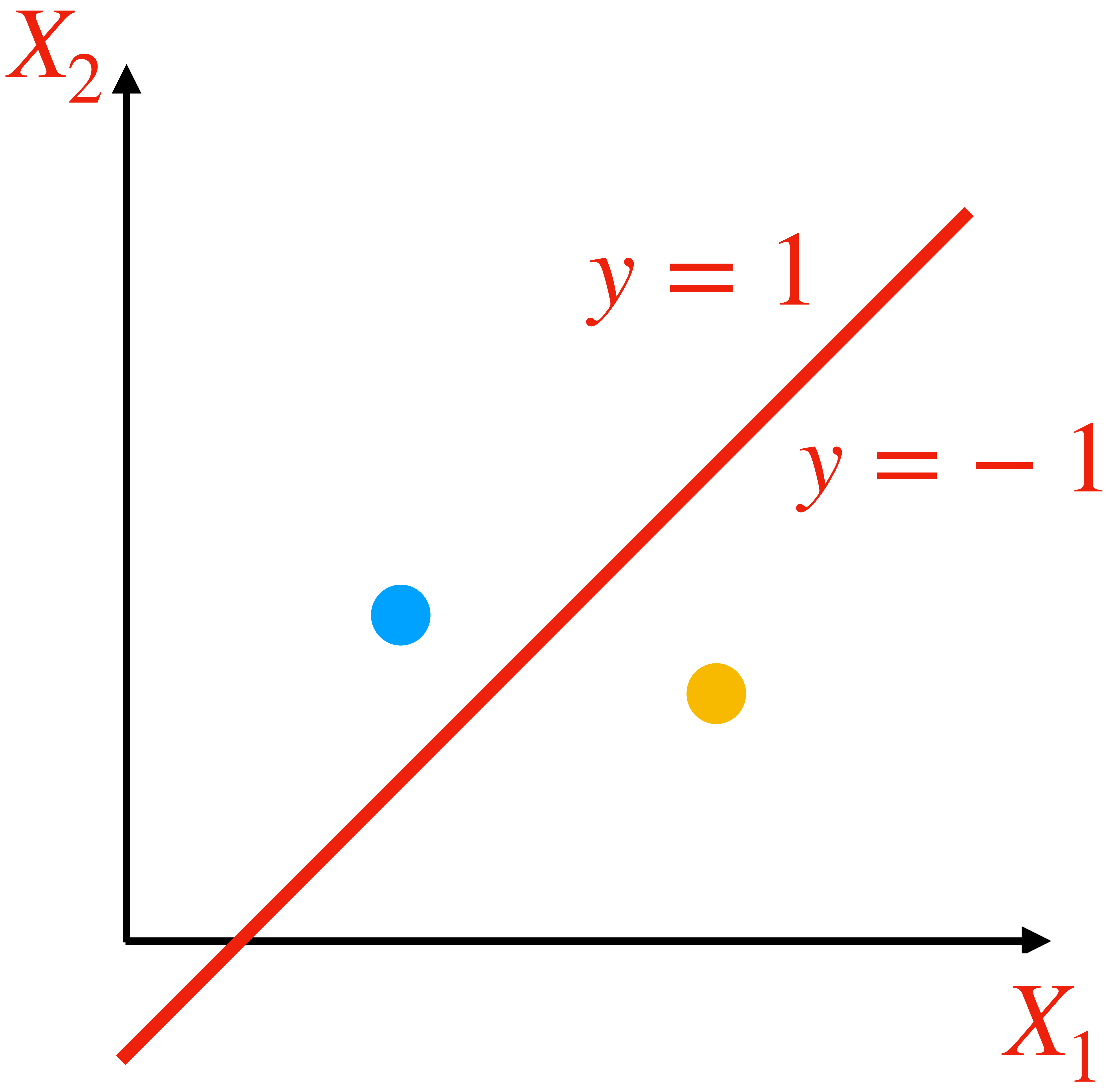

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For a training point i with $y_i = -1$:

$$w_0 + w_1 x_{i1} + w_2 x_{i2} < 0$$

Can combine them in one inequality:

$$y_i(w_0 + w_1 x_{i1} + w_2 x_{i2}) > 0$$

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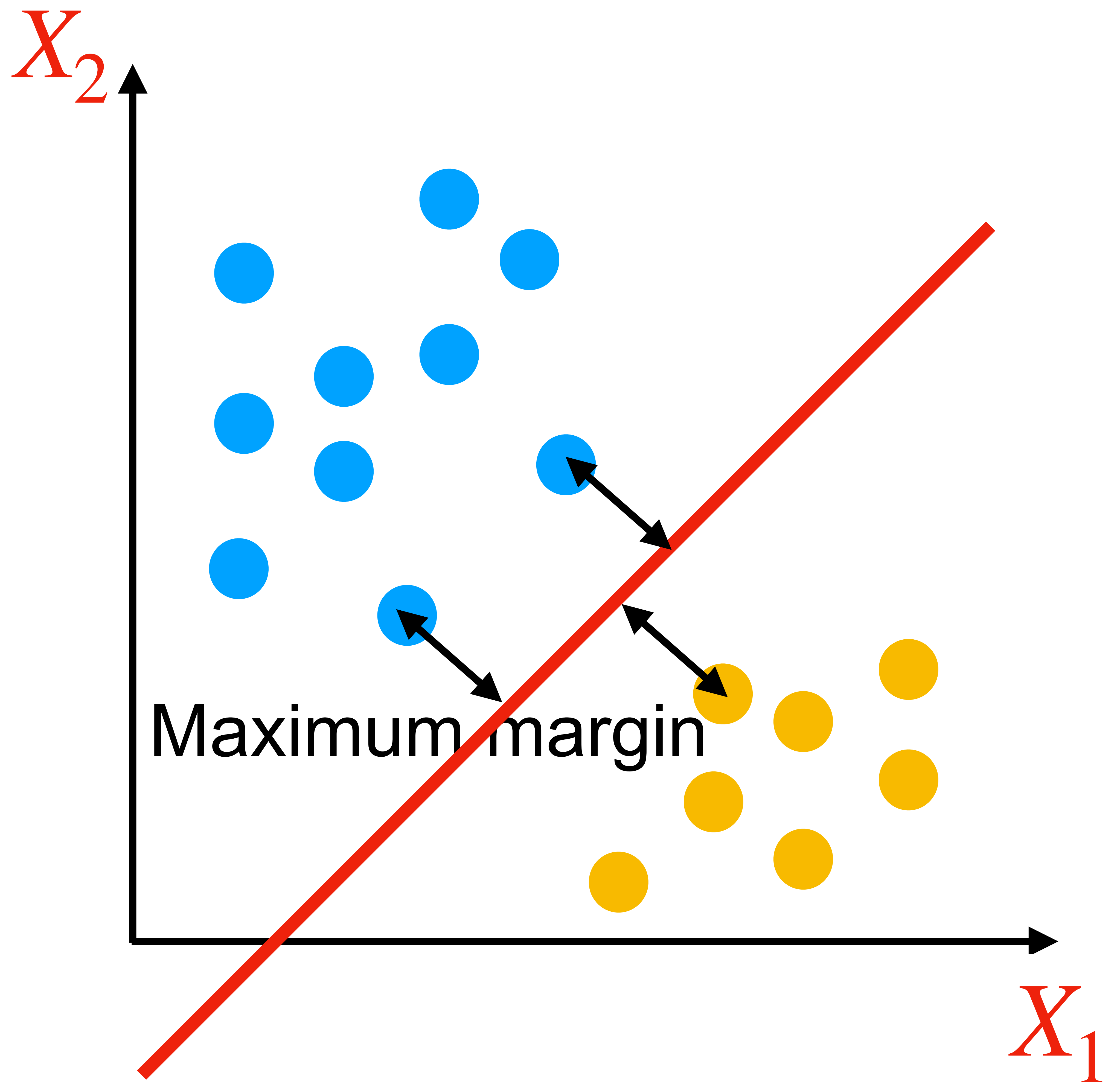


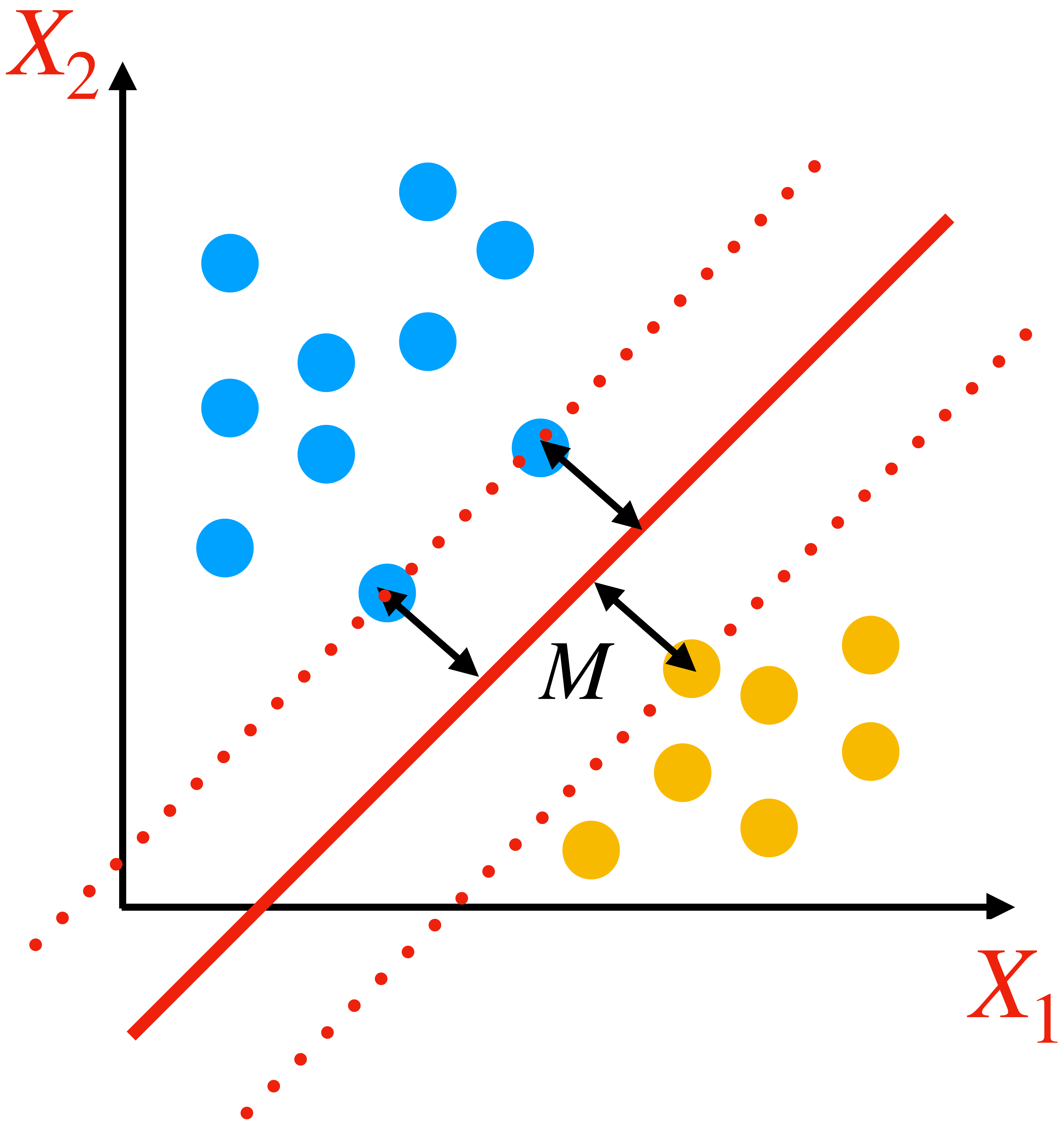
How about 'separates classes <u>well</u>':

Among red, green, purple:
red line separates the classes best.

Margin: minimum distance from any training point to the hyperplane.

Red line has the maximum margin.

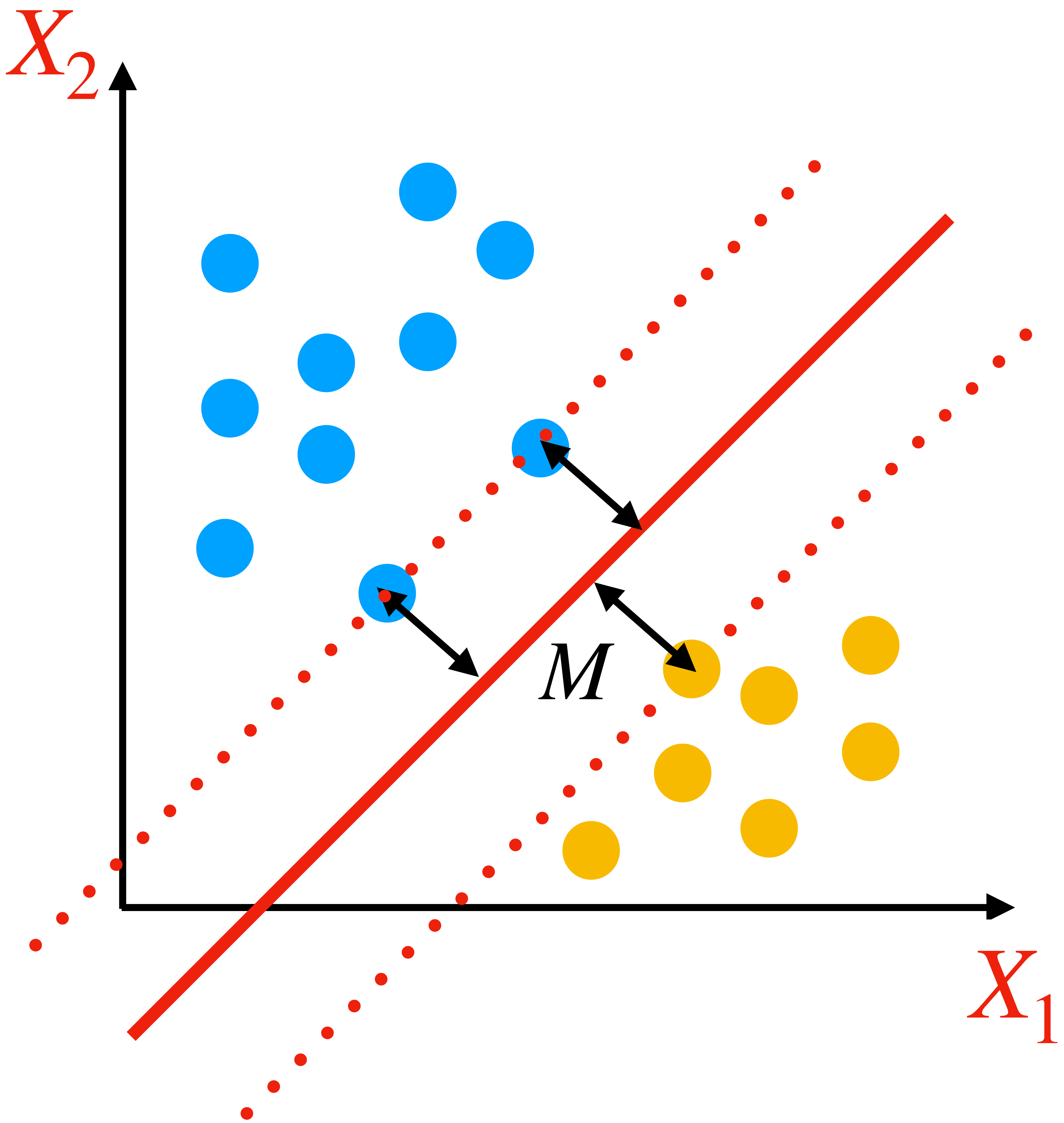
REVIEW OF PREVIOUS LECTURE: SUPPORT VECTOR MACHINES



Informally the goal is to:

Find hyperplane that provides maximum margin M.

REVIEW OF PREVIOUS LECTURE: SUPPORT VECTOR MACHINES



Formally the goal is to:

Find w_0, w_1, w_2 that

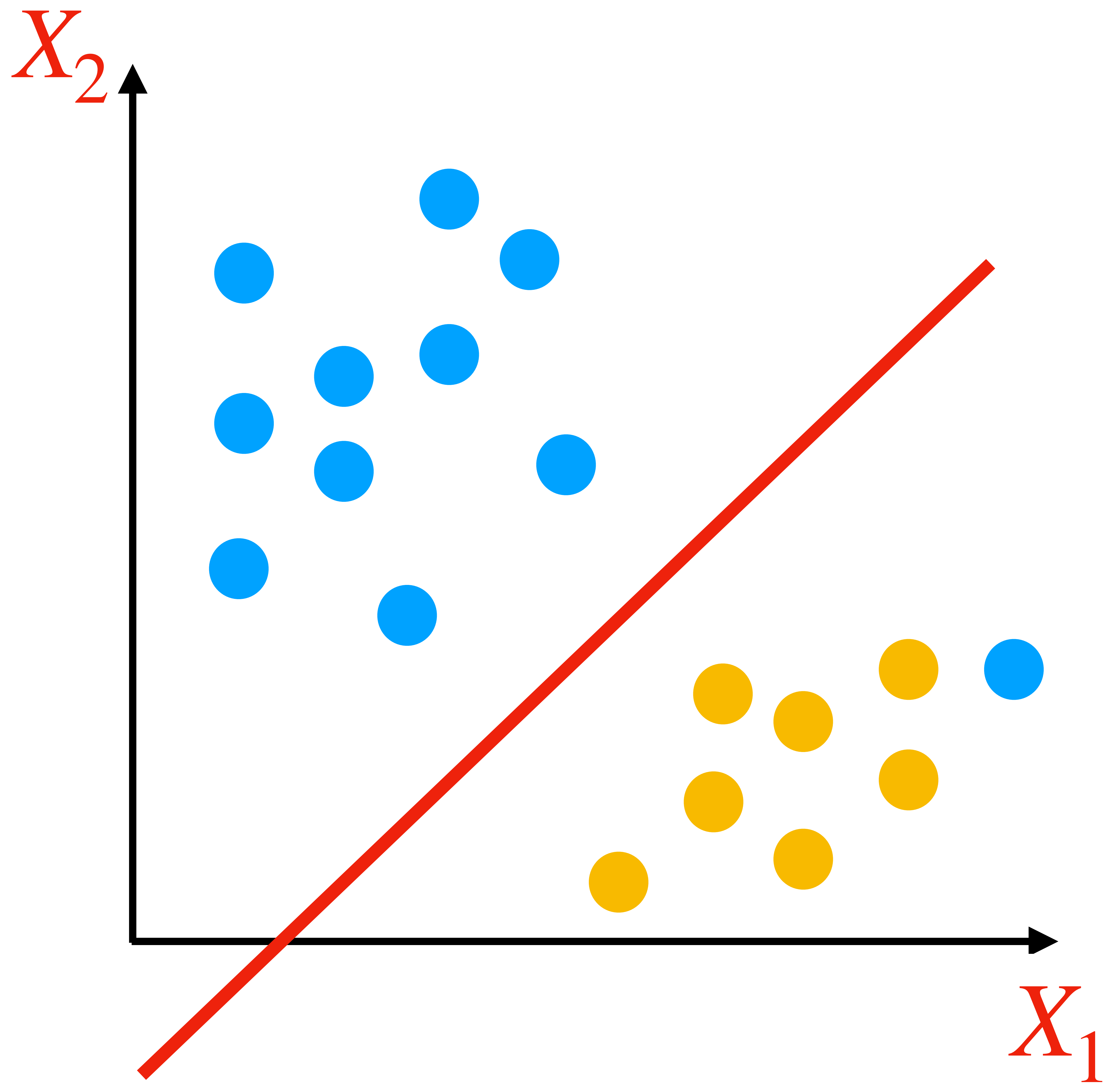
maximizes M

with the constraints

$\mathscr{C}_1$ $\boxed{w_1^2 + w_2^2 = 1}$

$\mathscr{C}_2$ $y_i(w_0 + w_1 x_{i1} + w_2 x_{i2}) \geq M$
for any training point i

REVIEW OF PREVIOUS LECTURE: SUPPORT VECTOR MACHINES



Two problems with the same solution:

Addition of a single point might drastically change the hyperplane.

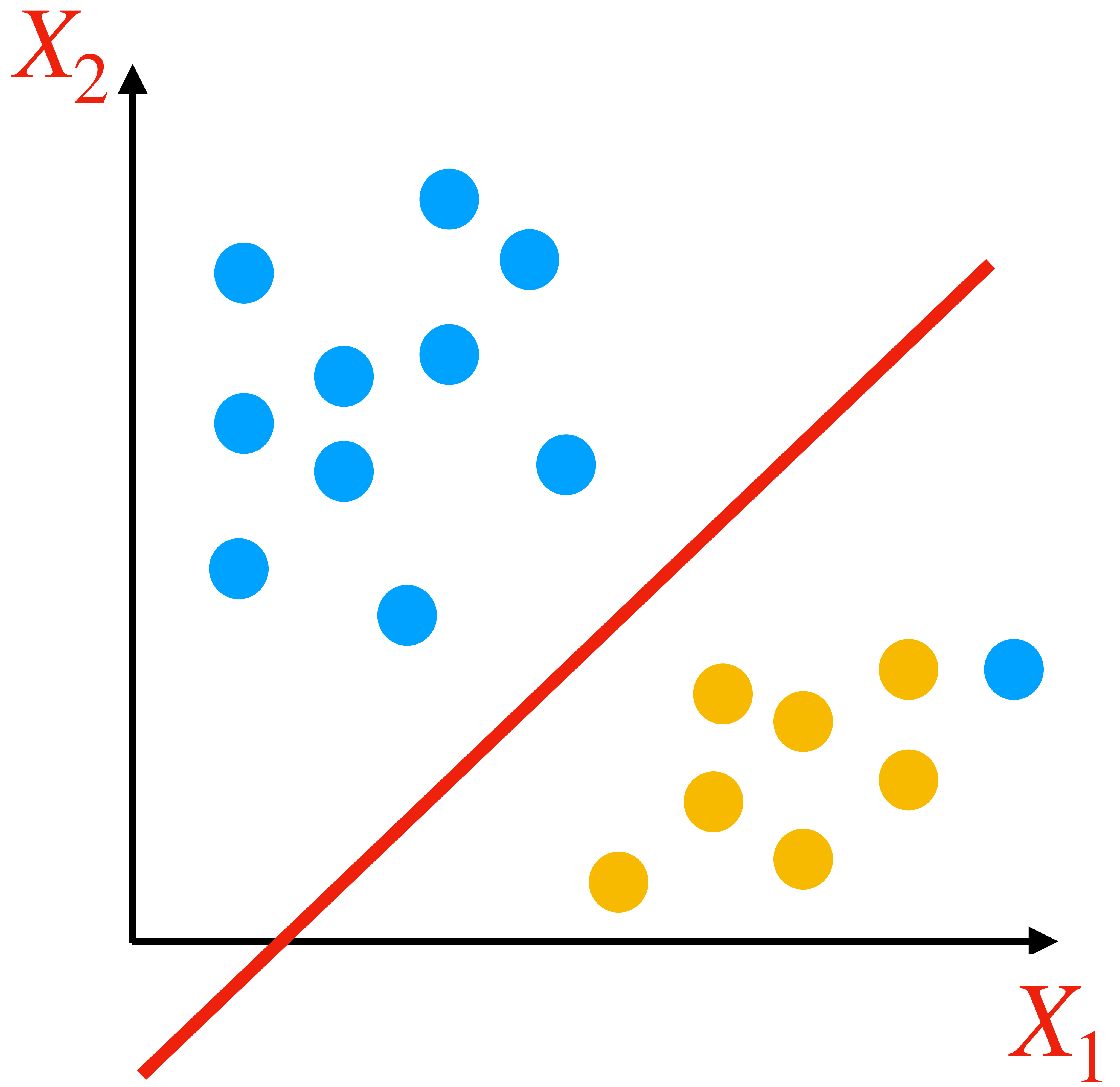
Insisting on perfect separation may cause overfitting!

It may not even be possible to find a separating hyperplane (line in 2D).

Solution is to use Soft Margin:

Add flexibility; some observations can be on the wrong side.

REVIEW OF PREVIOUS LECTURE: SUPPORT VECTOR MACHINES



Soft Margin:

Find w_0, w_1, w_2 that

maximizes M

with the constraints

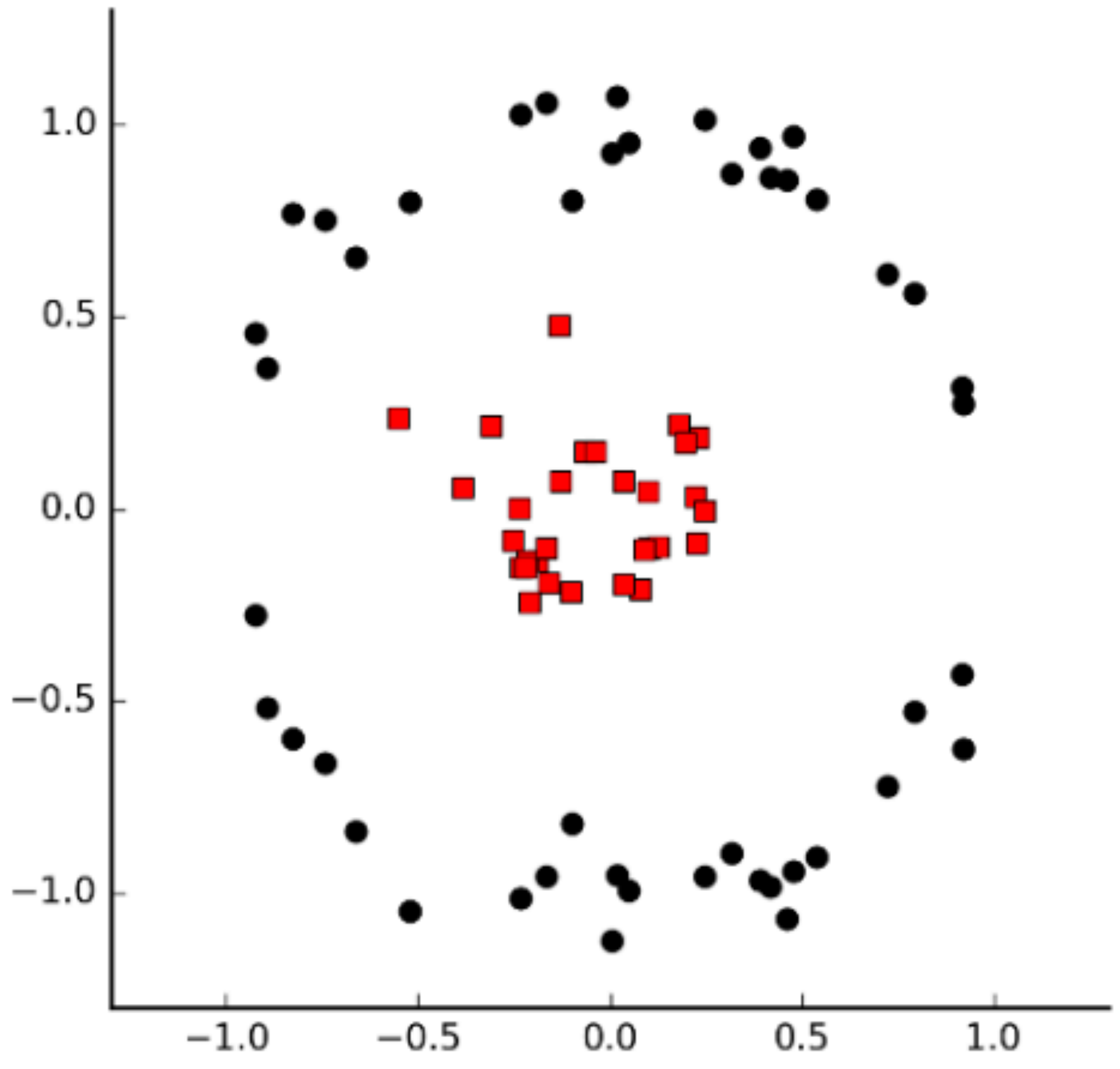
$\mathscr{C}_1$ $\boxed{w_1^2 + w_2^2 = 1}$

$\mathscr{C}_2$ $y_i(w_0 + w_1 x_{i1} + w_2 x_{i2}) \geq M(1 - \epsilon_i)$ for any training point i

$\mathscr{C}_3$ $\boxed{\epsilon_i \geq 0, \sum_{i=1}^{n} \epsilon_i \leq C}$

SUPPORT VECTOR MACHINES (SVM)

What if we need nonlinear separation?



SUPPORT VECTOR MACHINES (SVM)

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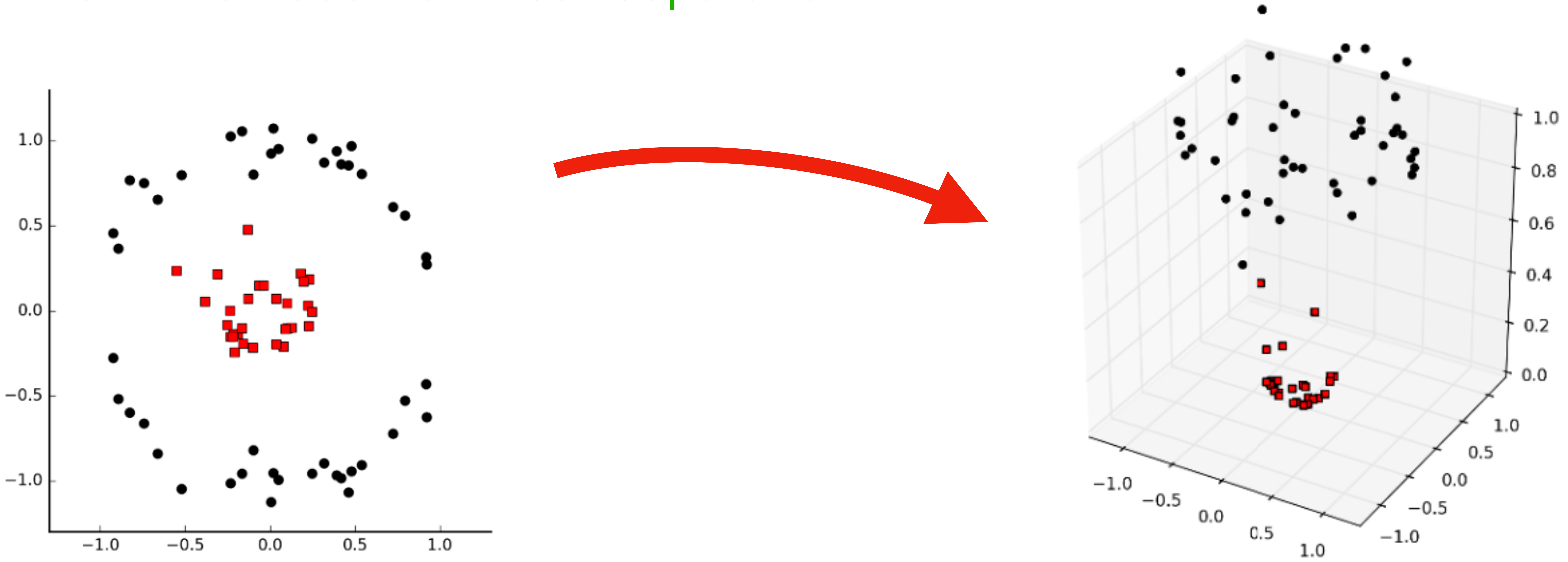
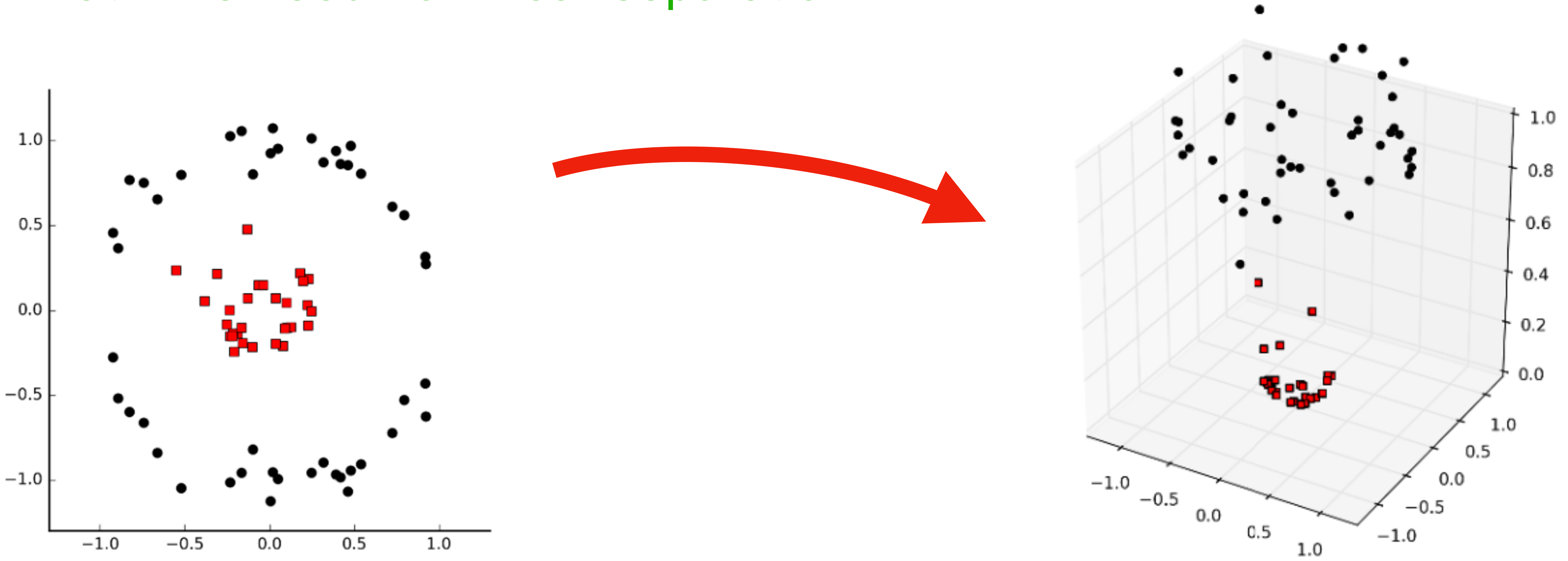


Project m dimensional points to higher dimensions.

Ex: $(x, y) \rightarrow (x, y, x^2 + y^2)$

SUPPORT VECTOR MACHINES (SVM)

What if we need nonlinear separation?



How to project points each of dimension m into points of dimension n?

Kernel functions

SUPPORT VECTOR MACHINES (SVM)

Kernel Functions

Normally for a training point (x_1, x_2) we mainly check the function

$w_0 + w_1x_1 + w_2x_2$ (If positive class 1, otherwise class -1)

SUPPORT VECTOR MACHINES (SVM)

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This is equivalent to checking the function:

$$w_0 + \sum_{i=1}^{n} \alpha_i(x \cdot x_i)$$

Dot product

Recall dot product measures similarity (linear relationship).

SUPPORT VECTOR MACHINES (SVM)

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This is equivalent to checking the function:

$$w_0 + \sum_{i=1}^{n} \alpha_i \underbrace{(x \cdot x_i)}_{\text{Dot product}}$$

Recall dot product measures similarity (linear relationship).

Generalize: Replace dot product with any appropriate similarity measure, called kernel

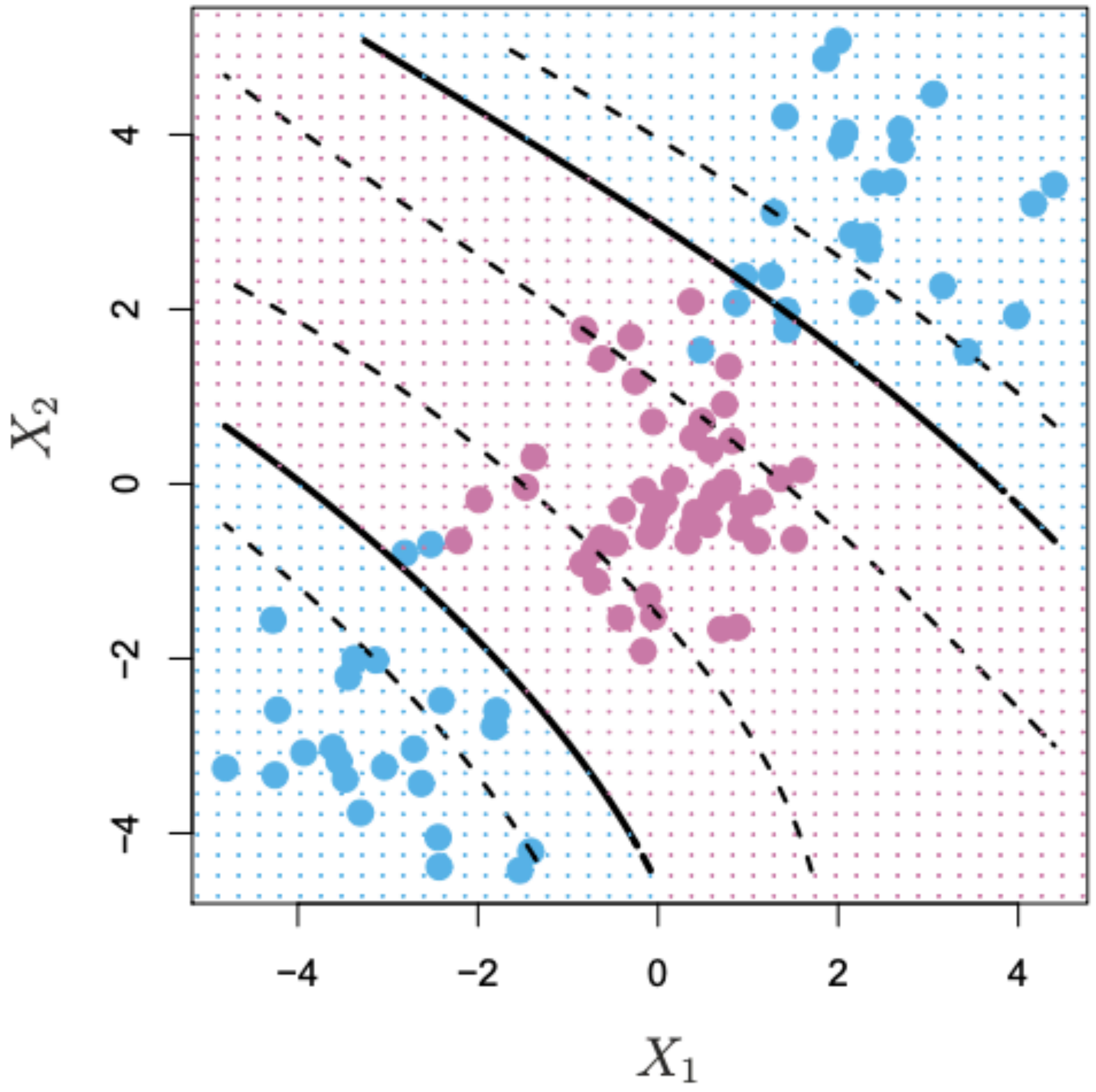
$$w_0 + \sum_{i=1}^{n} \alpha_i \underbrace{K(x, x_i)}_{\text{Kernel}}$$

Popular non-linear kernels: Polynomial, radial

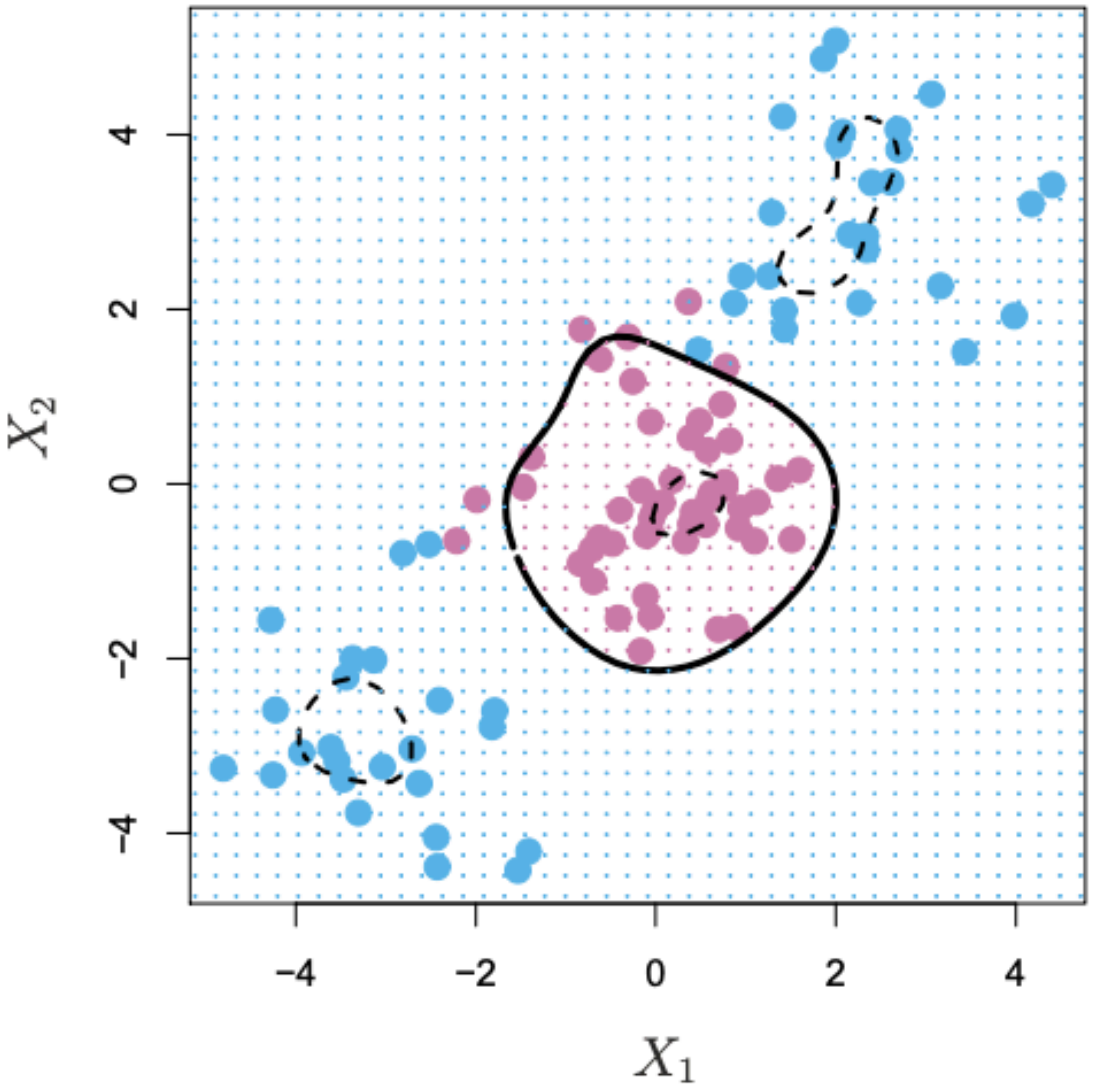
SUPPORT VECTOR MACHINES (SVM)

Kernel Functions

Example: Non-linear data



Polynomial ($d = 3$) kernel



Radial kernel

SUPPORT VECTOR MACHINES (SVM)

SVM in scikit learn:

```
class sklearn.svm.SVC(*, C=1.0, kernel='rbf', degree=3, gamma='scale', coef0=0.0,
shrinking=True, probability=False, tol=0.001, cache_size=200, class_weight=None,
verbose=False, max_iter=-1, decision_function_shape='ovr', break_ties=False,
random_state=None)
```

Regularization parameter.

Note: strength of regularization is inversely proportional to C.

(penalty of the error term. small C implies wider margin - may misclassify more points)

Kernel type:

linear, poly, rbf (radial), etc.

DECISION TREES

Ex: Simple decision tree for predicting survival in Titanic

Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

DECISION TREES

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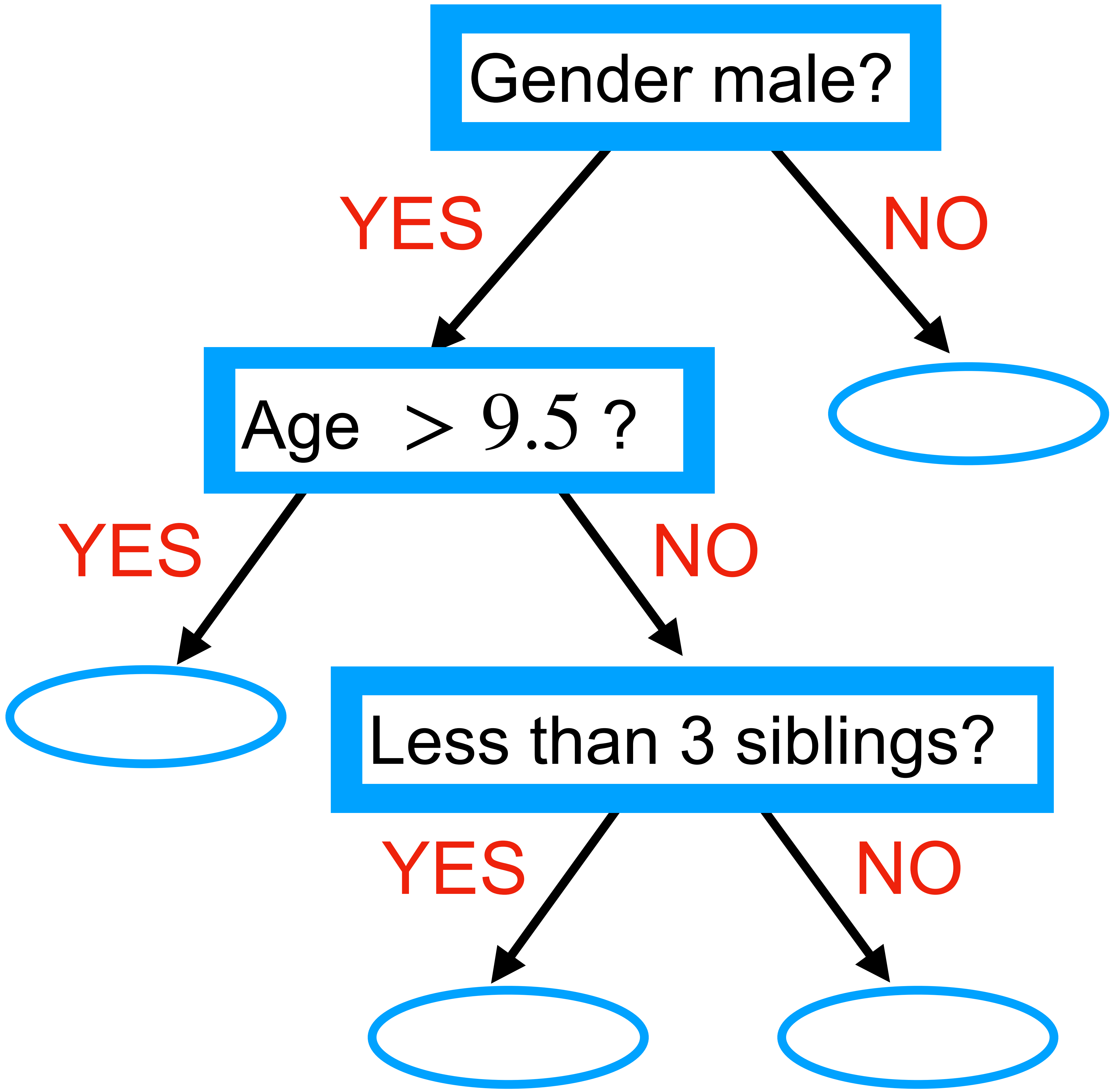

Each node corresponds to a group of training data points.

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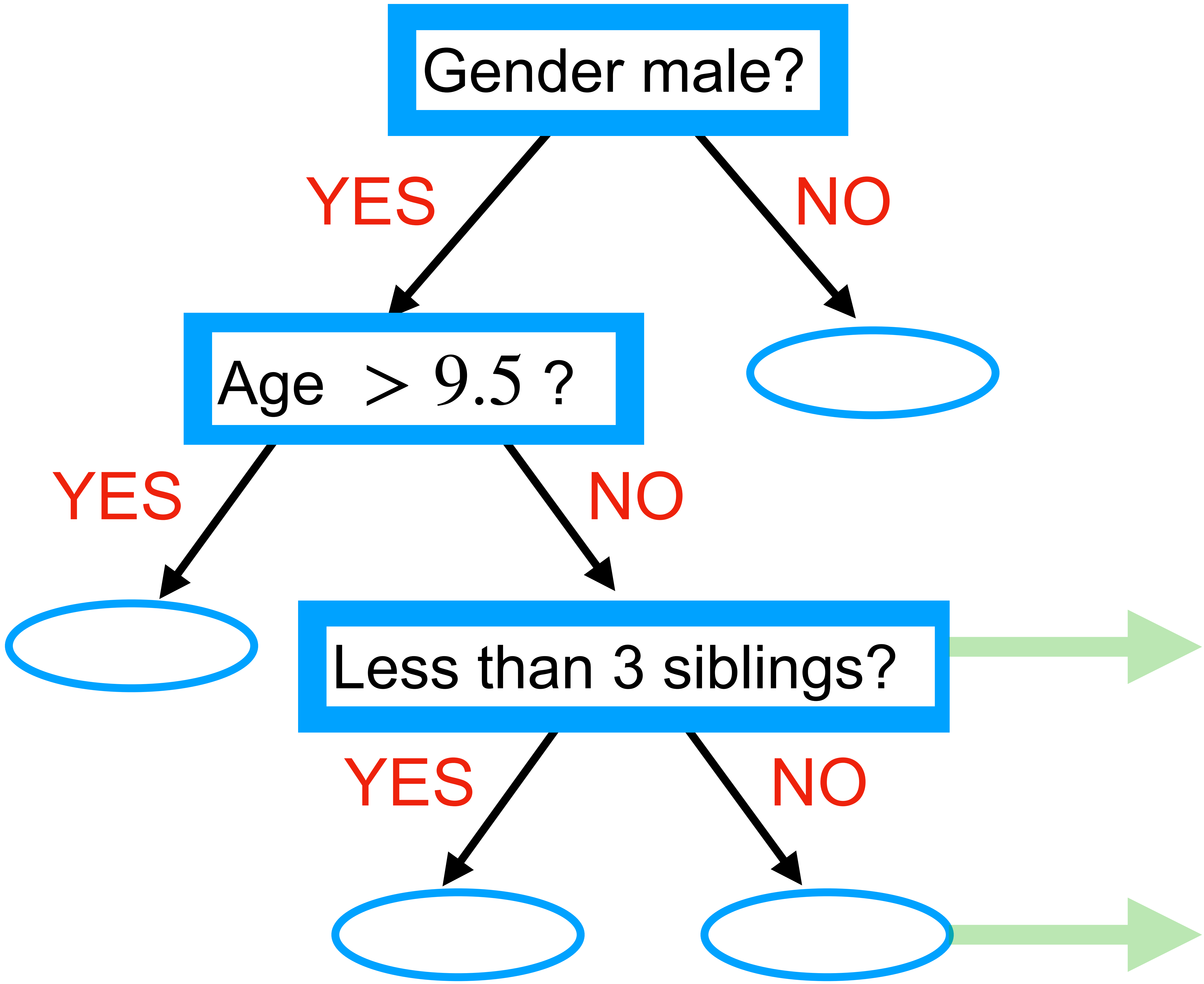

Each node corresponds to a group of training data points.

Example:

Passengers who are male and ≤ 9.5 years old.

Passengers who are male and ≤ 9.5 years old and have more than 2 siblings.

DECISION TREES

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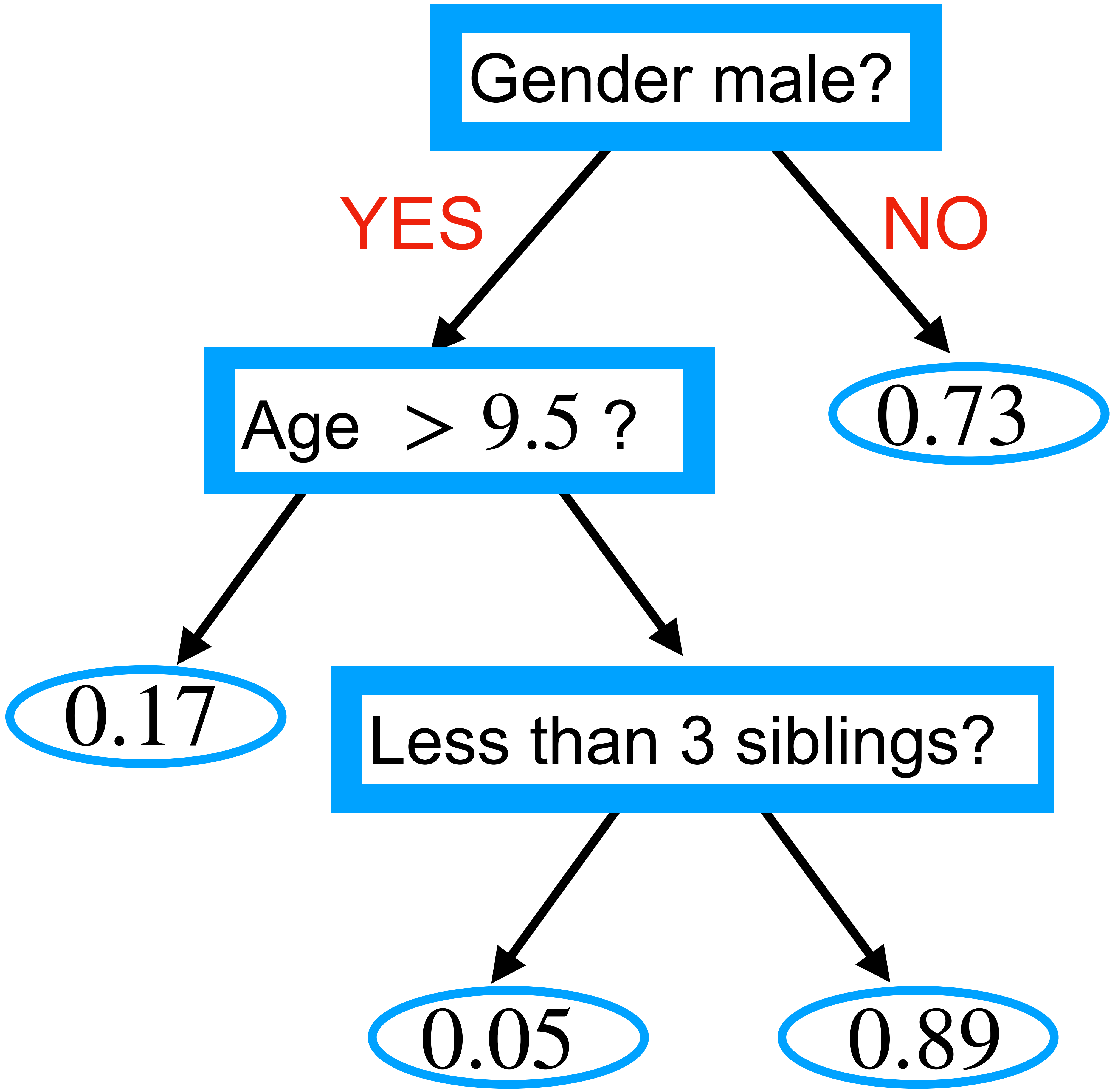

For classifying a new test point:

Start at the root of the tree.

Traverse the tree until a terminal node.

Classify with the majority class.

(Number at a terminal node is the fraction of survivors in that group)

DECISION TREES

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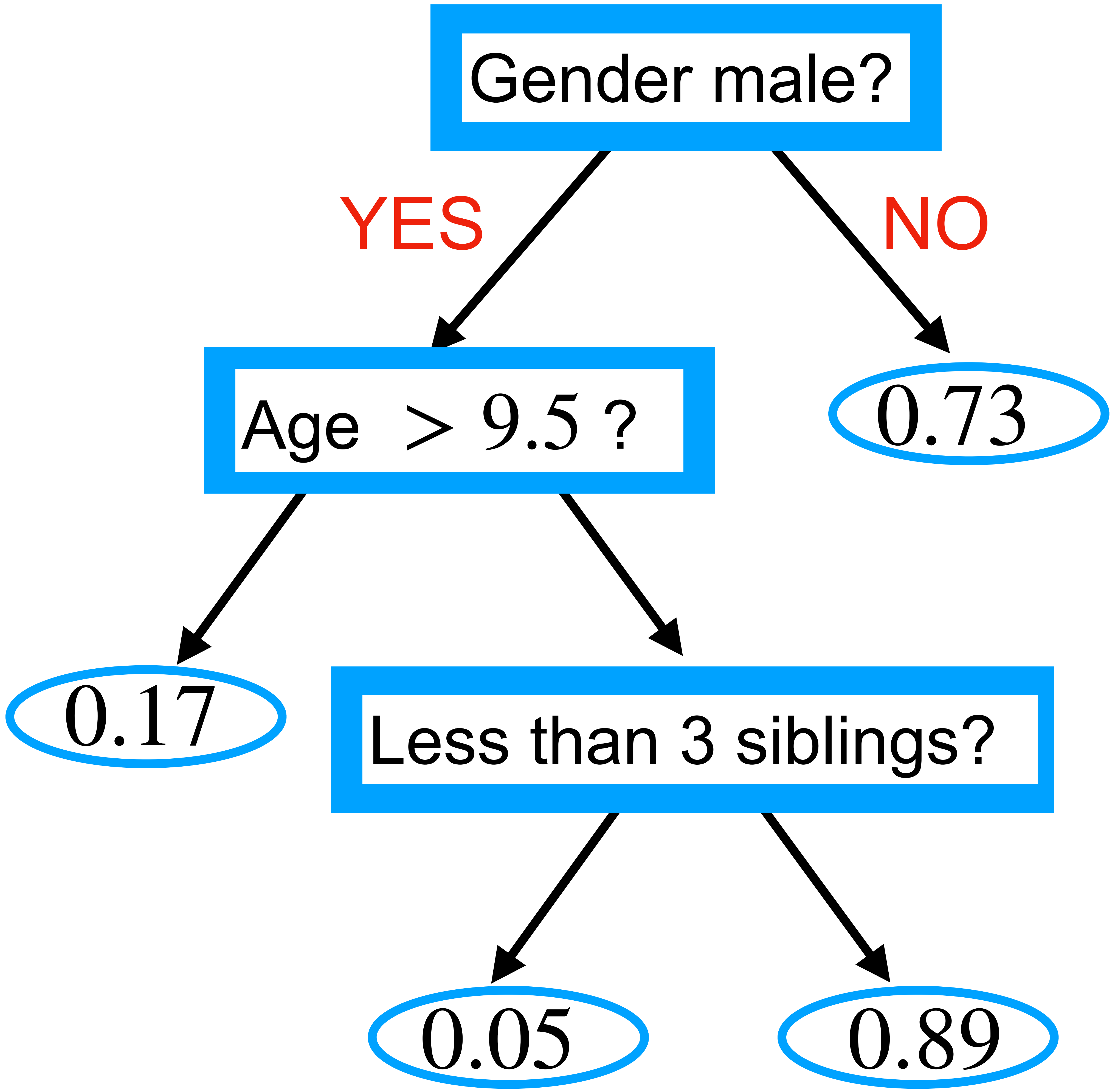

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Male, 8 years old, with one sibling?

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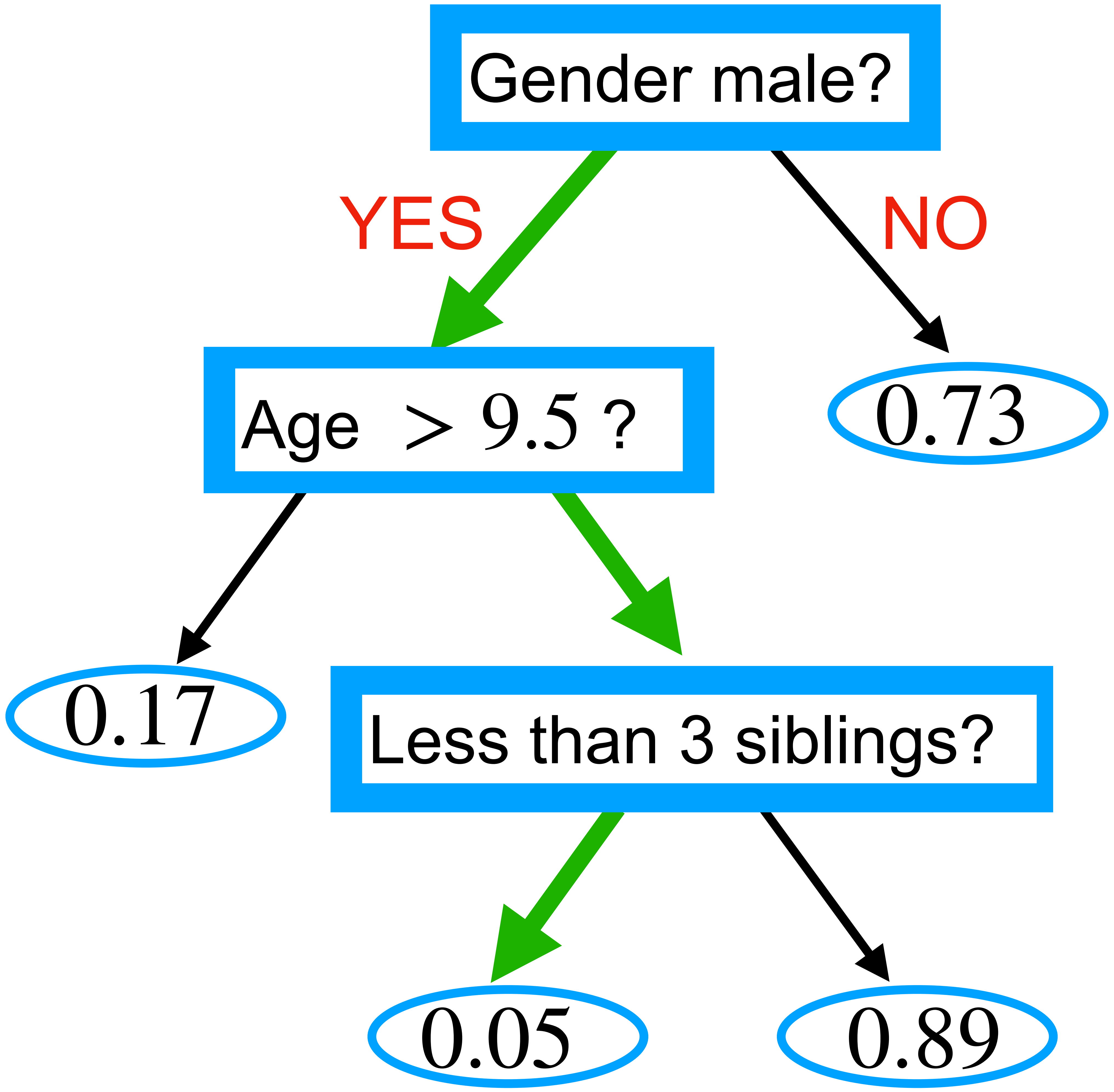

Majority class in group is 'Not survived'.

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Start at the root of the tree.

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Example:

Male, 8 years old, with one sibling?

Not survived.

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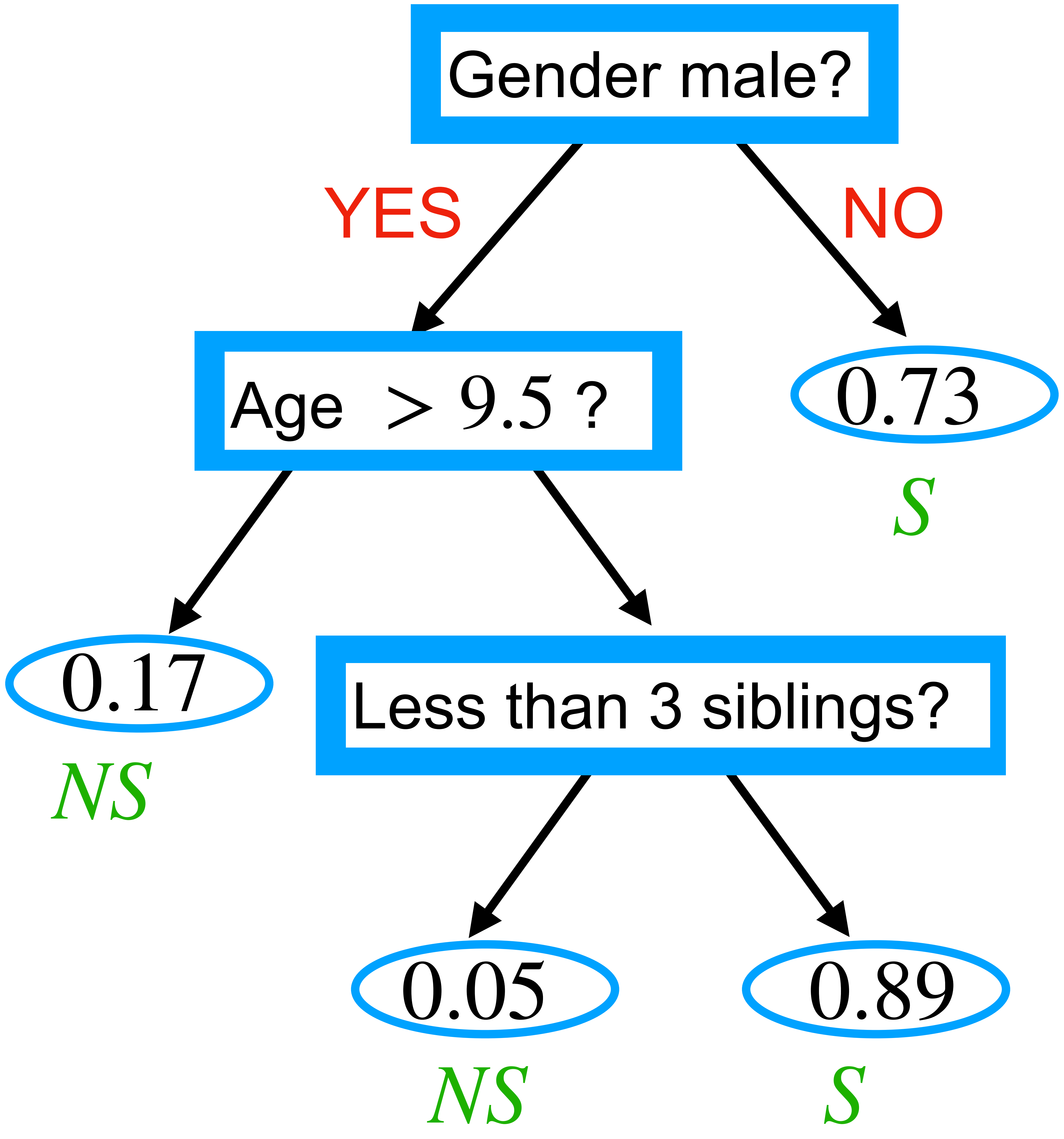
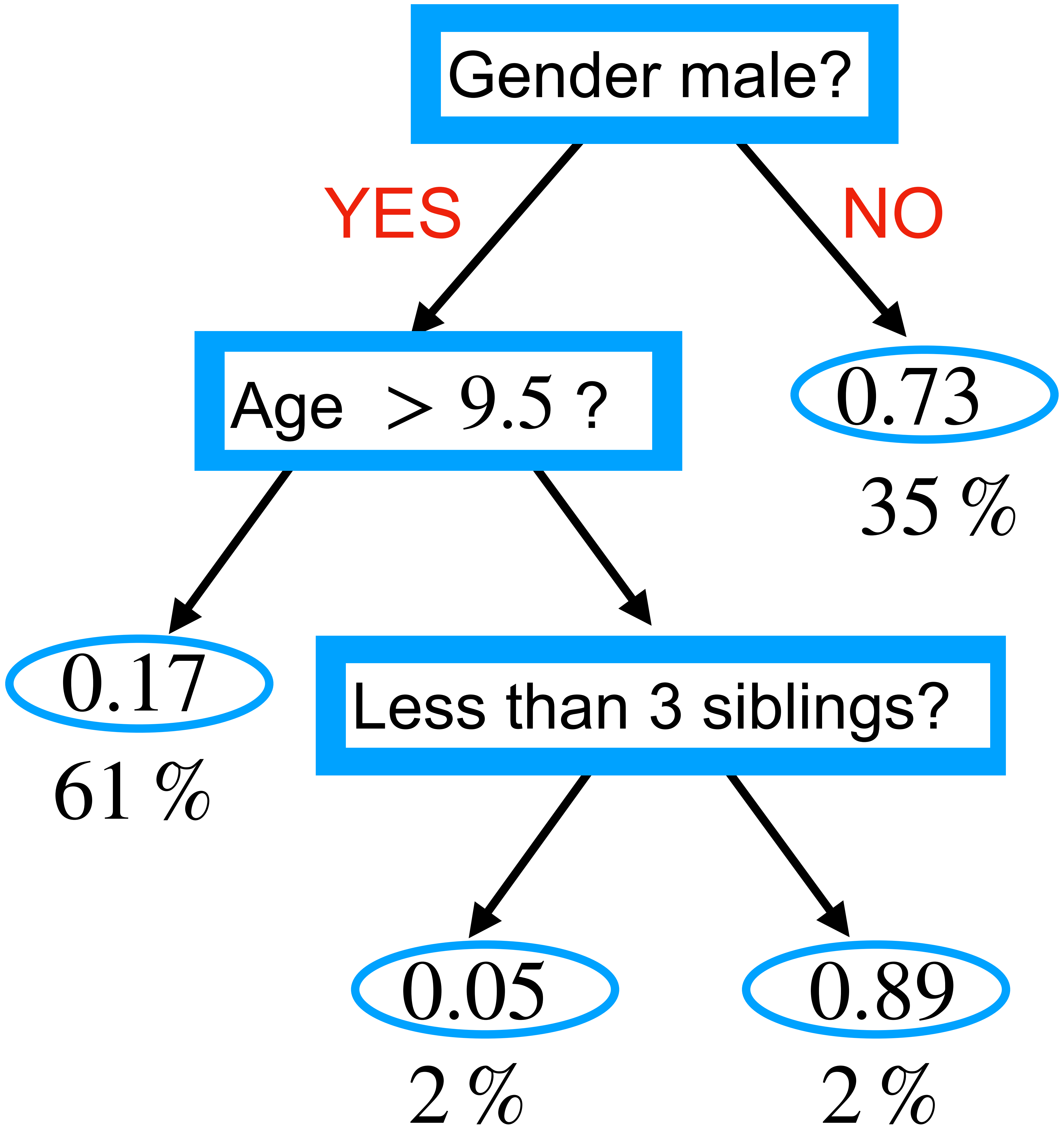


Classification of each point falling into a terminal node would be as shown.
(S: Survived, NS: not survived)

DECISION TREES

Ex: Simple decision tree for predicting survival in Titanic



Training accuracy of the decision tree?

(Percentage at a terminal node is the percentage of the group in Titanic.)

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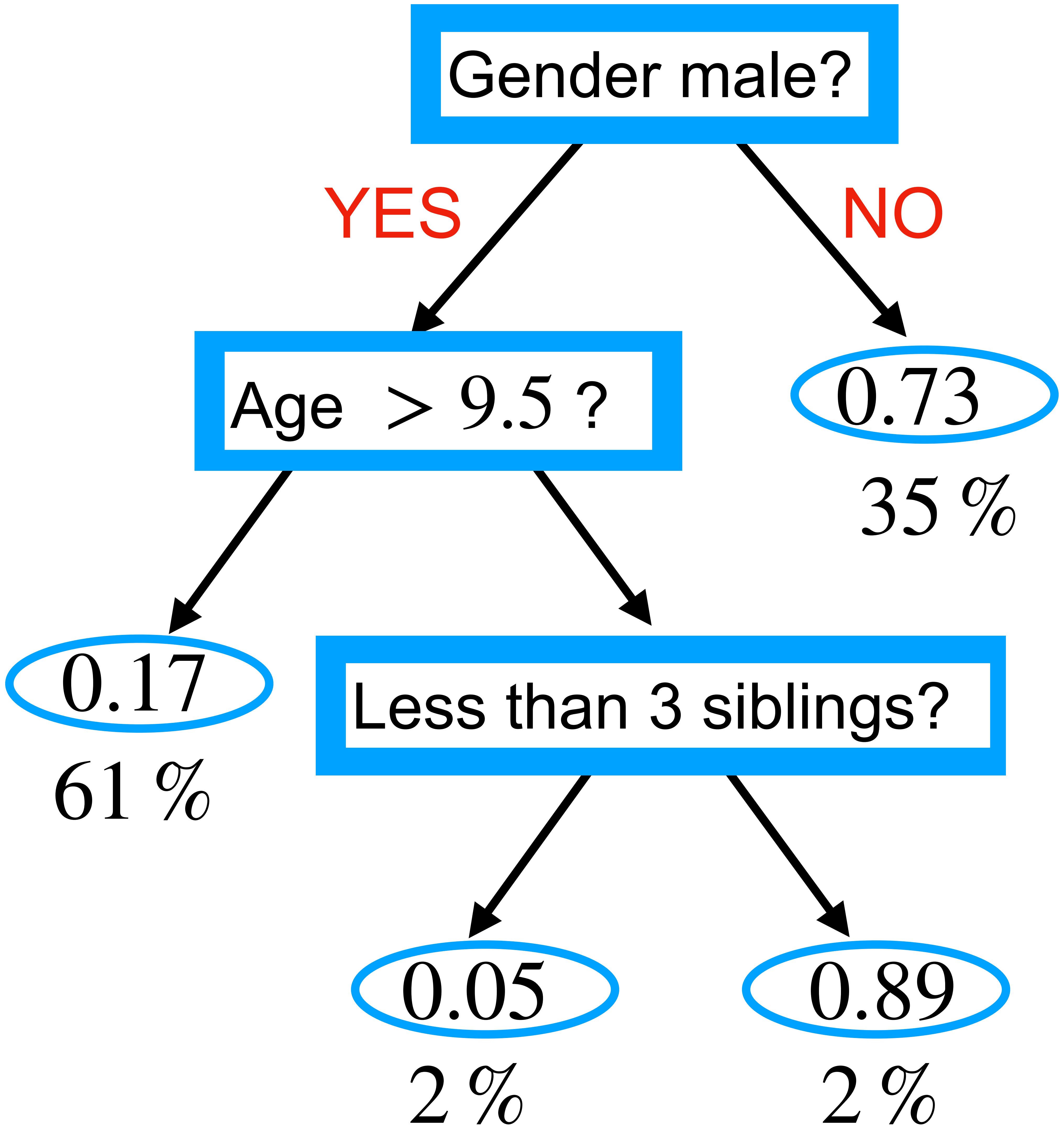

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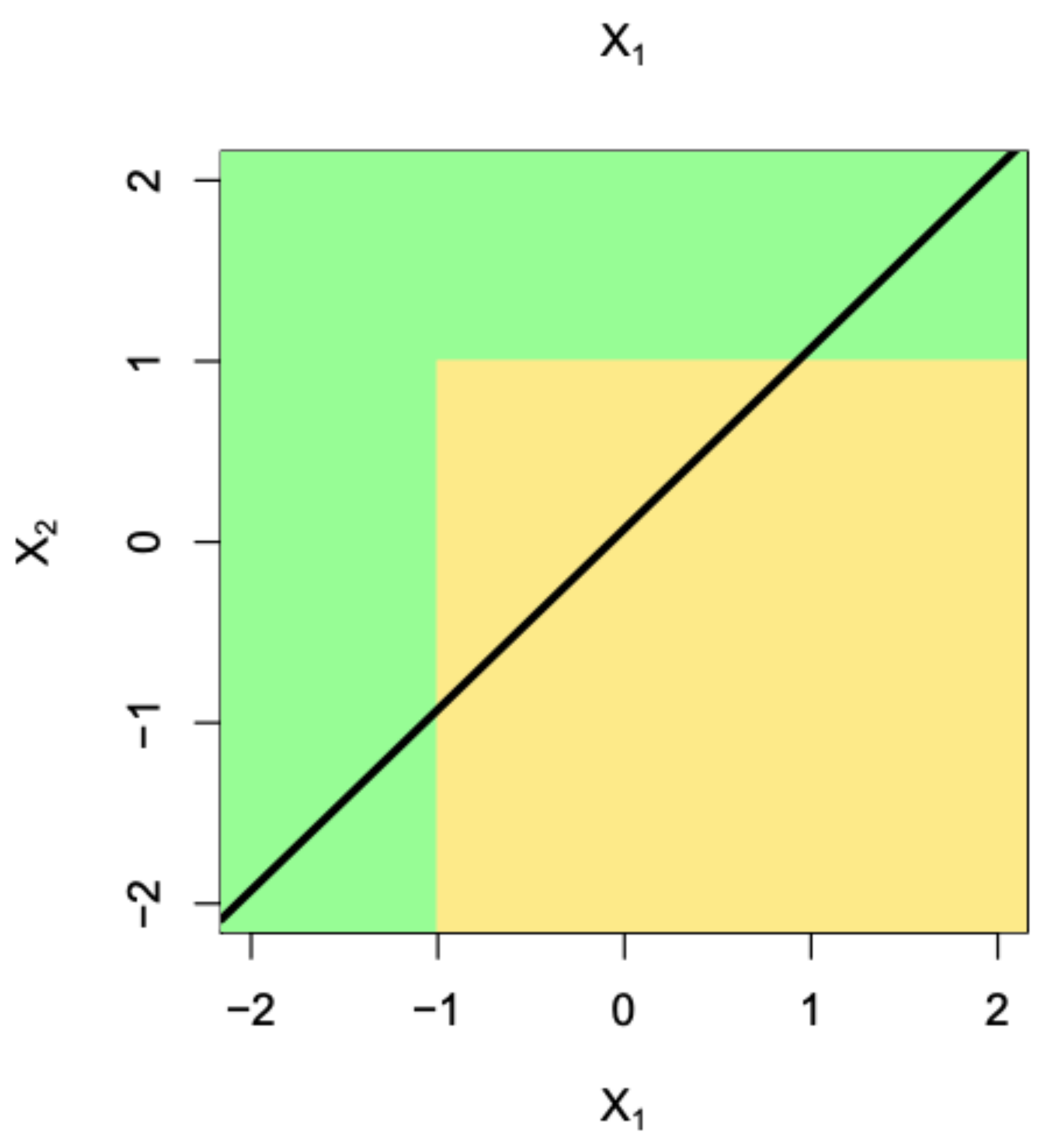
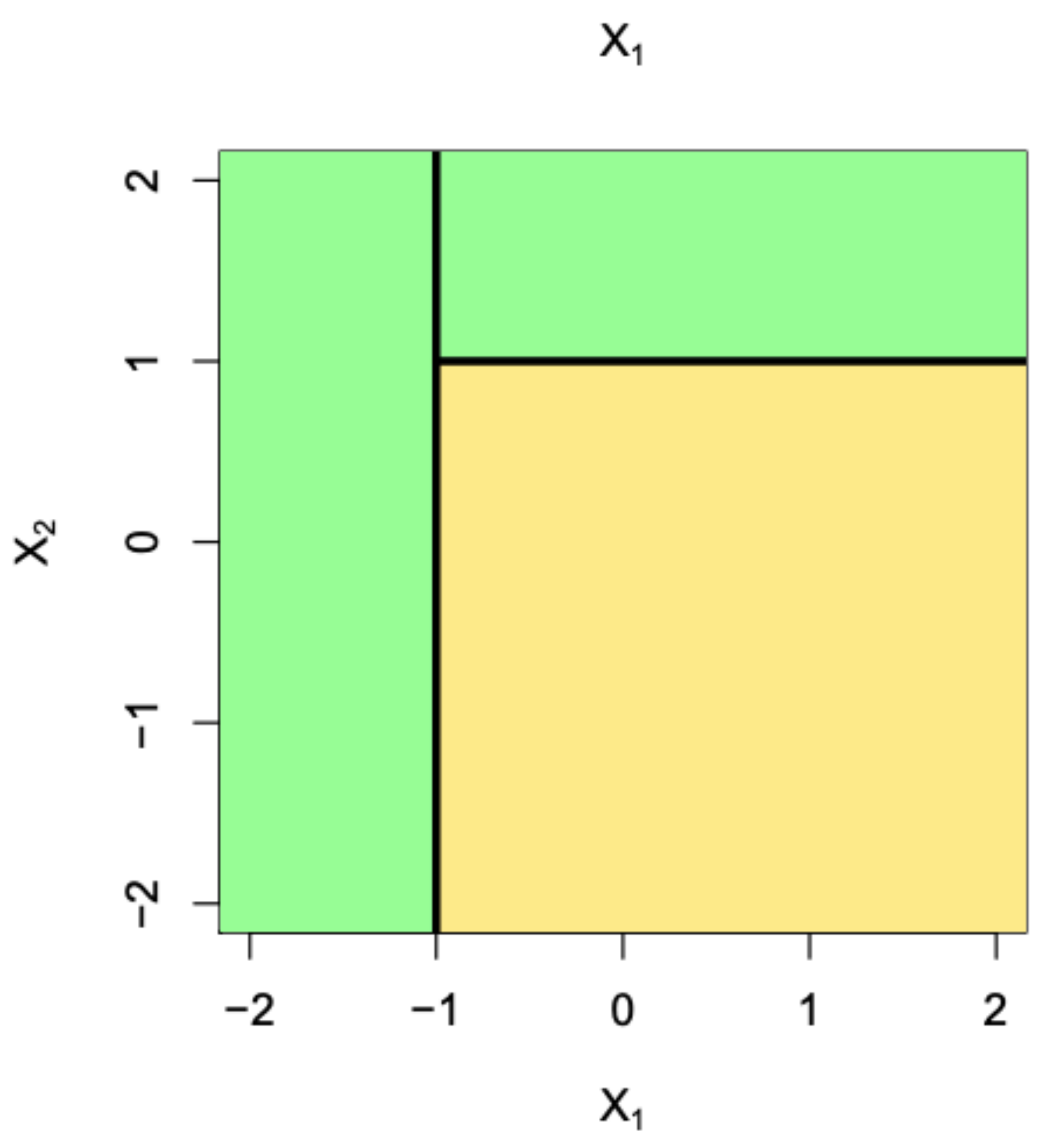
$$0.73 \times 0.35 + 0.83 \times 0.61 + 0.95 \times 0.02 + 0.89 \times 0.02$$

ADVANTAGES OF DECISION TREES

Non-linearity:



Linear model



Decision tree

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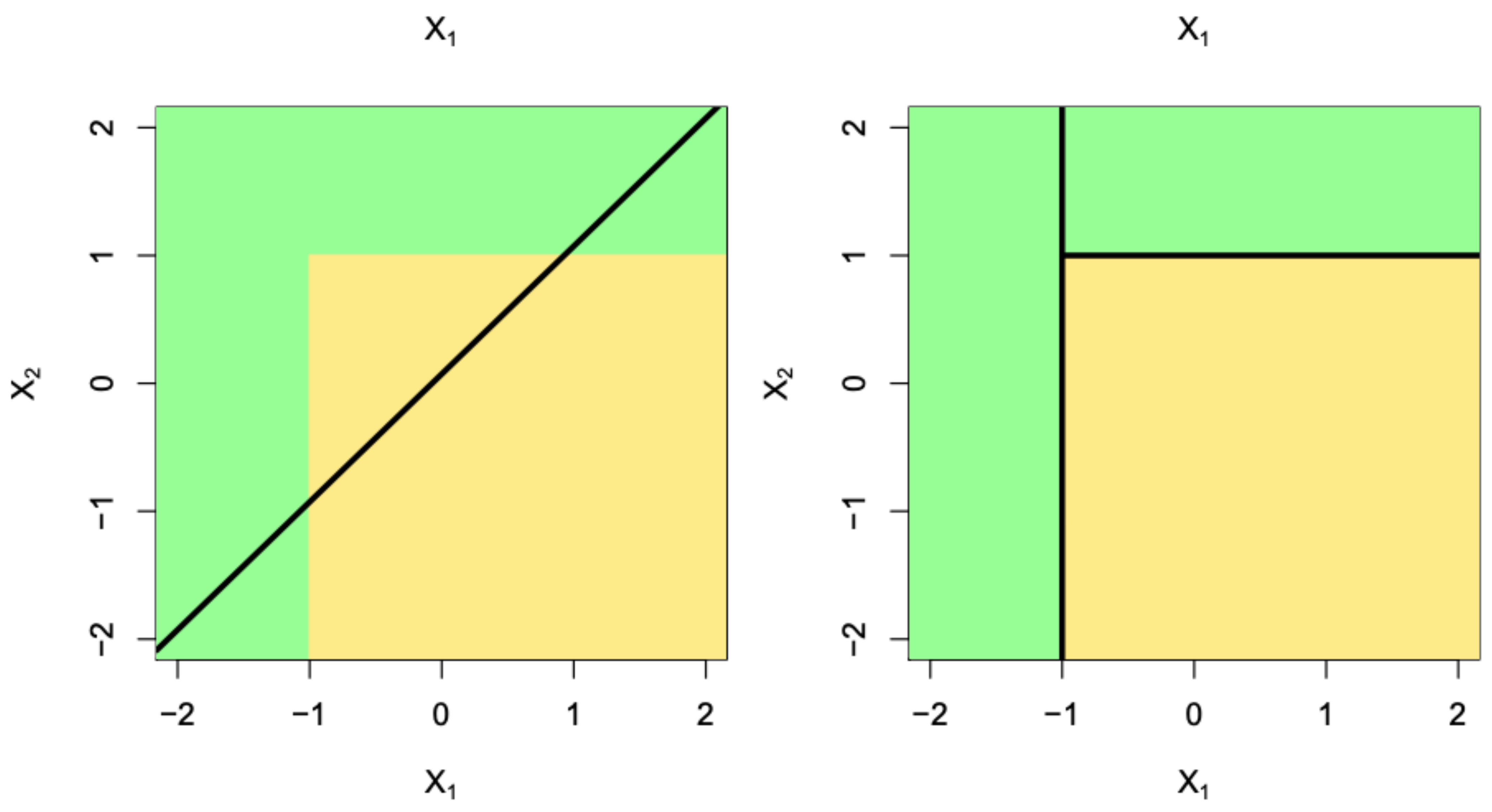

Linear model   Decision tree

Support for categorical variables (hair=red):

The splits of the tree can be done based on categorical values without the need for one hot encoding.

ADVANTAGES OF DECISION TREES

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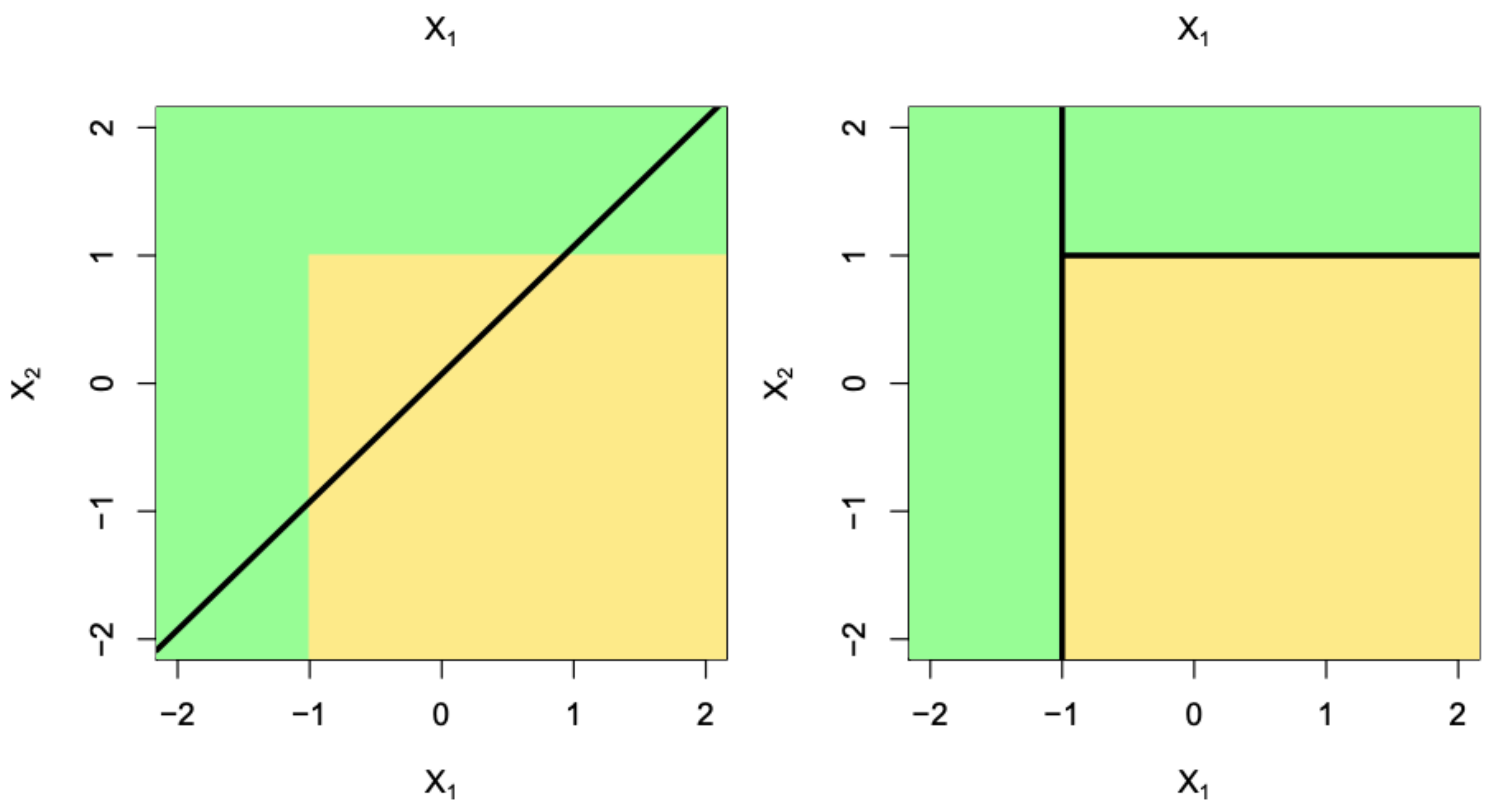

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Interpretability:

Features closest to root are important.

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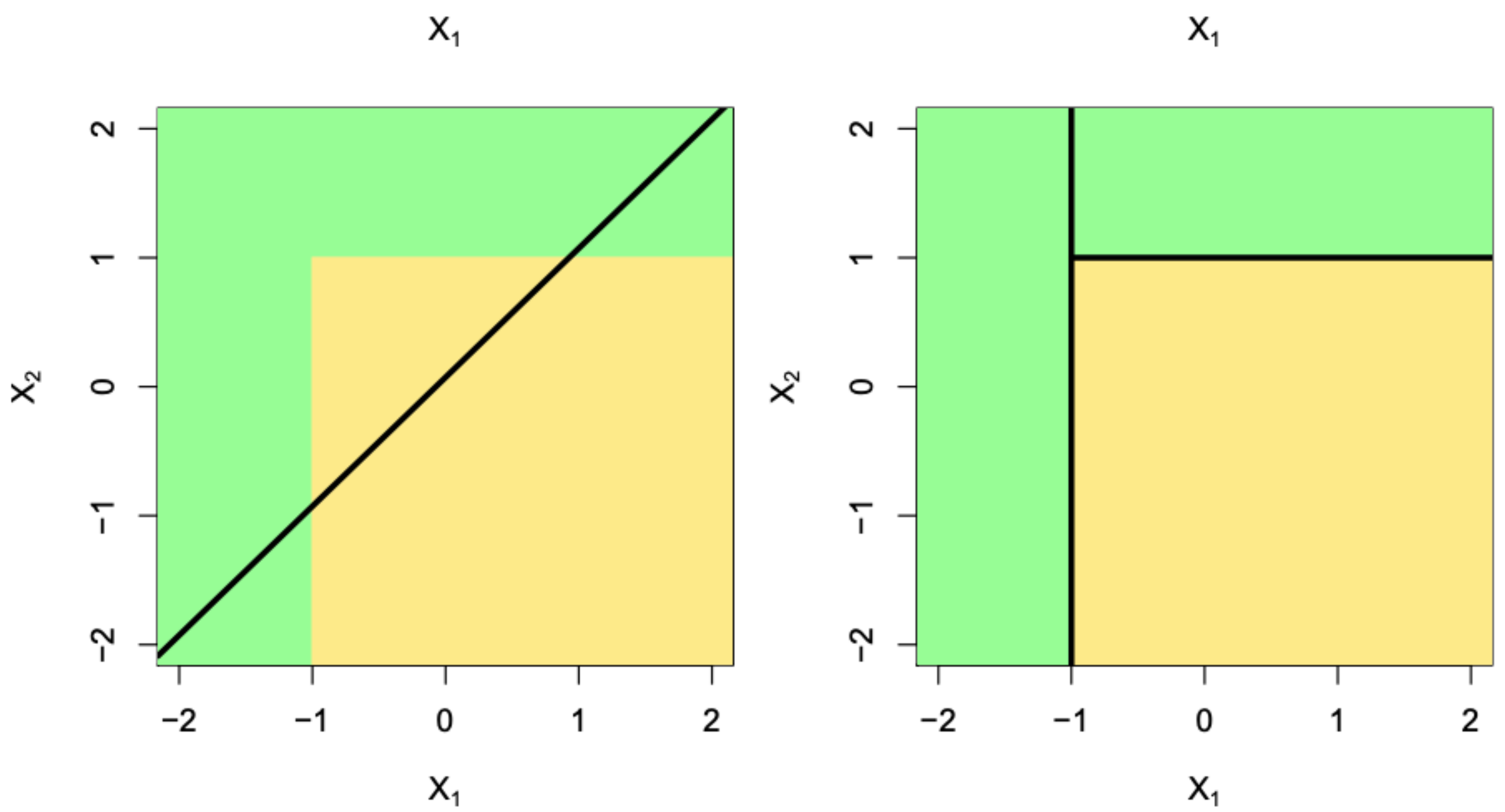

Linear model   Decision tree

Support for categorical variables (hair=red):

The splits of the tree can be done based on categorical values without the need for one hot encoding.

Interpretability:

Features closest to root are important.

Applies to regression:

Take the mean of the group at the terminal node.

HOW IS A DECISION TREE CONSTRUCTED?

Recursive binary splitting: Top-down, greedy.

- Each node: a binary predicate derived from a predictor..

- Discrete valued features: is $x_i = v$? where v: a possible value

  Numerical features: is $x_i \geq t$? where t: a threshold

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  Numerical features: is $x_i \geq t$? where t: a threshold

Important question:

Which predictor x_i and threshold t (or value v) to split at some node?

Try all predictors and all possible thresholds (basic decision tree).

Find the 'best' one.

HOW IS A DECISION TREE CONSTRUCTED?

Example: Two classes: Red or blue. Two predictors, x_1: numerical, x_2: categorical.

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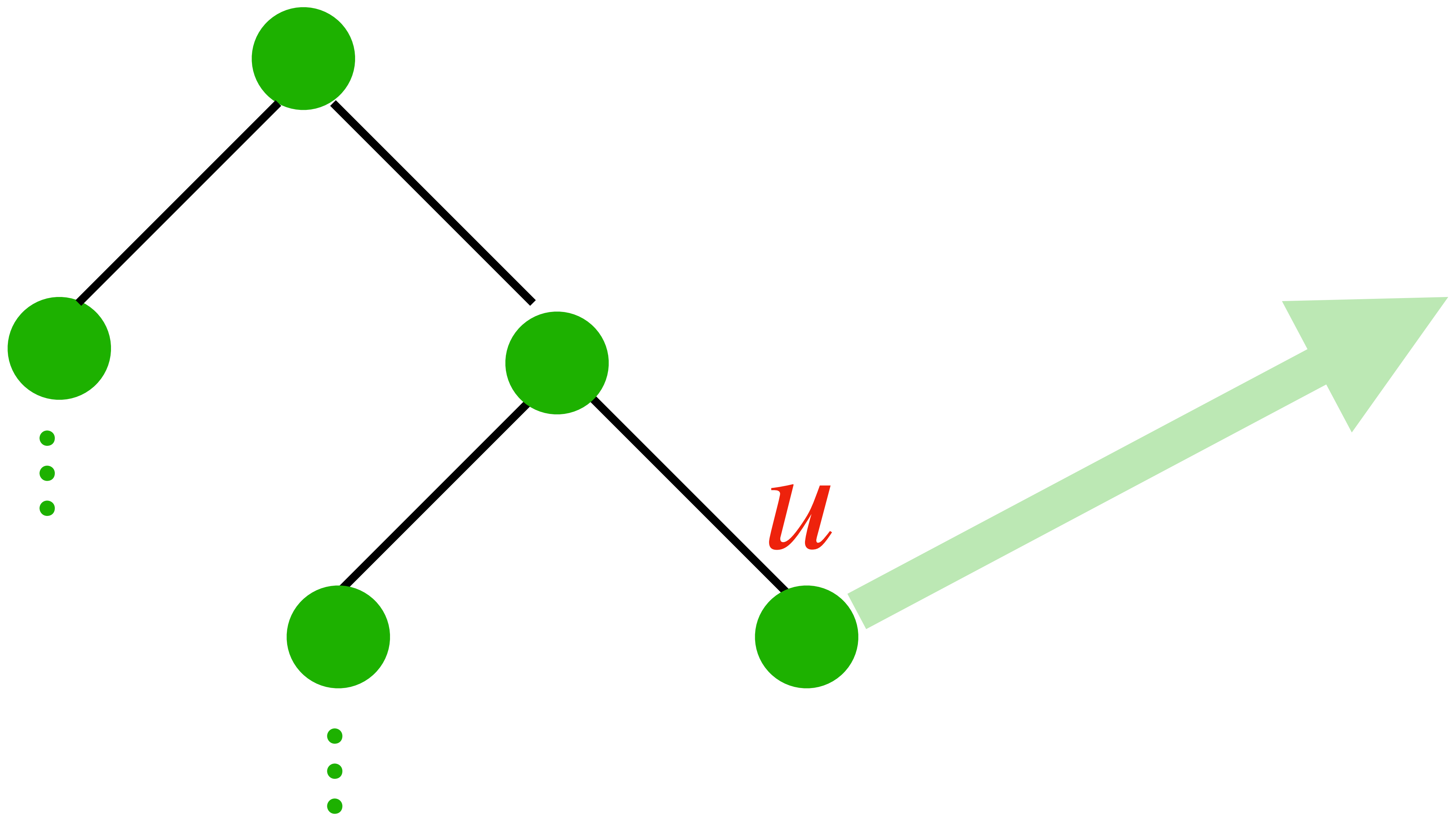
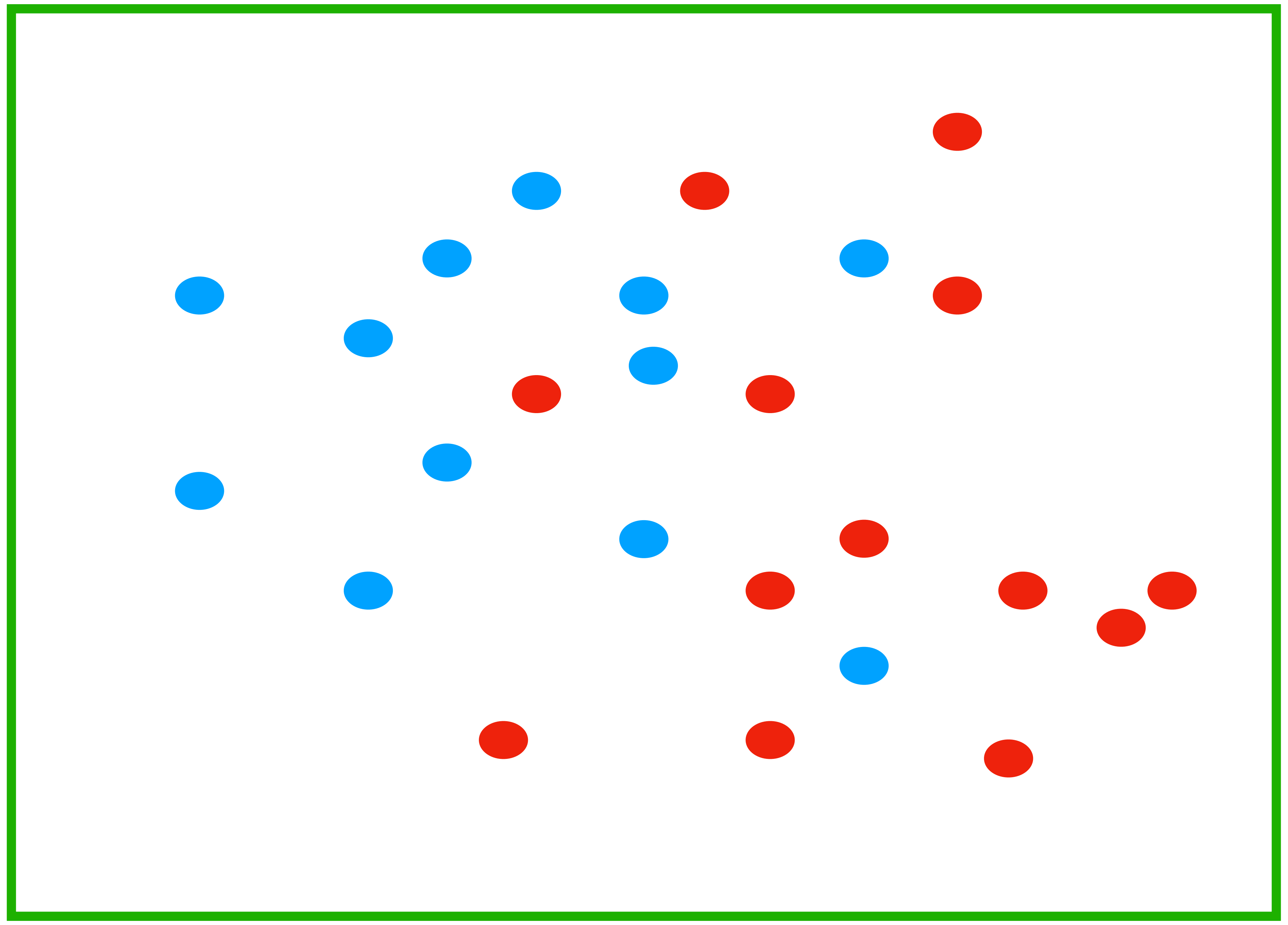


Decision tree constructed up to now and we are trying to do split at node u



Training points at node u

HOW IS A DECISION TREE CONSTRUCTED?

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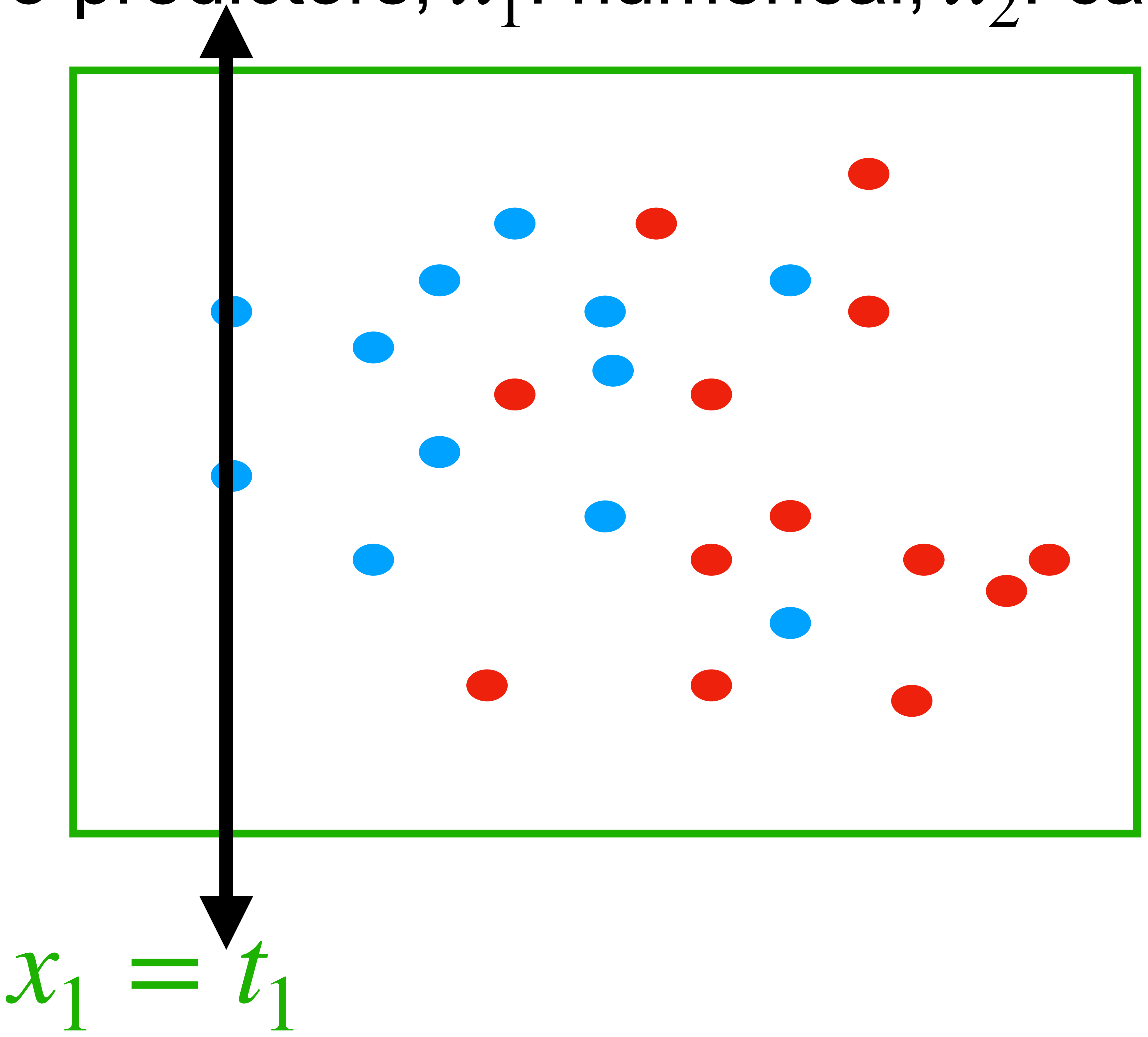

Boolean predicate at u?

Try all possible thresholds based on predictor x_1 and record 'quality of split' for each.

Example thresholds:

$x_1 \leq t_1$ vs $x_1 > t_1$ OR

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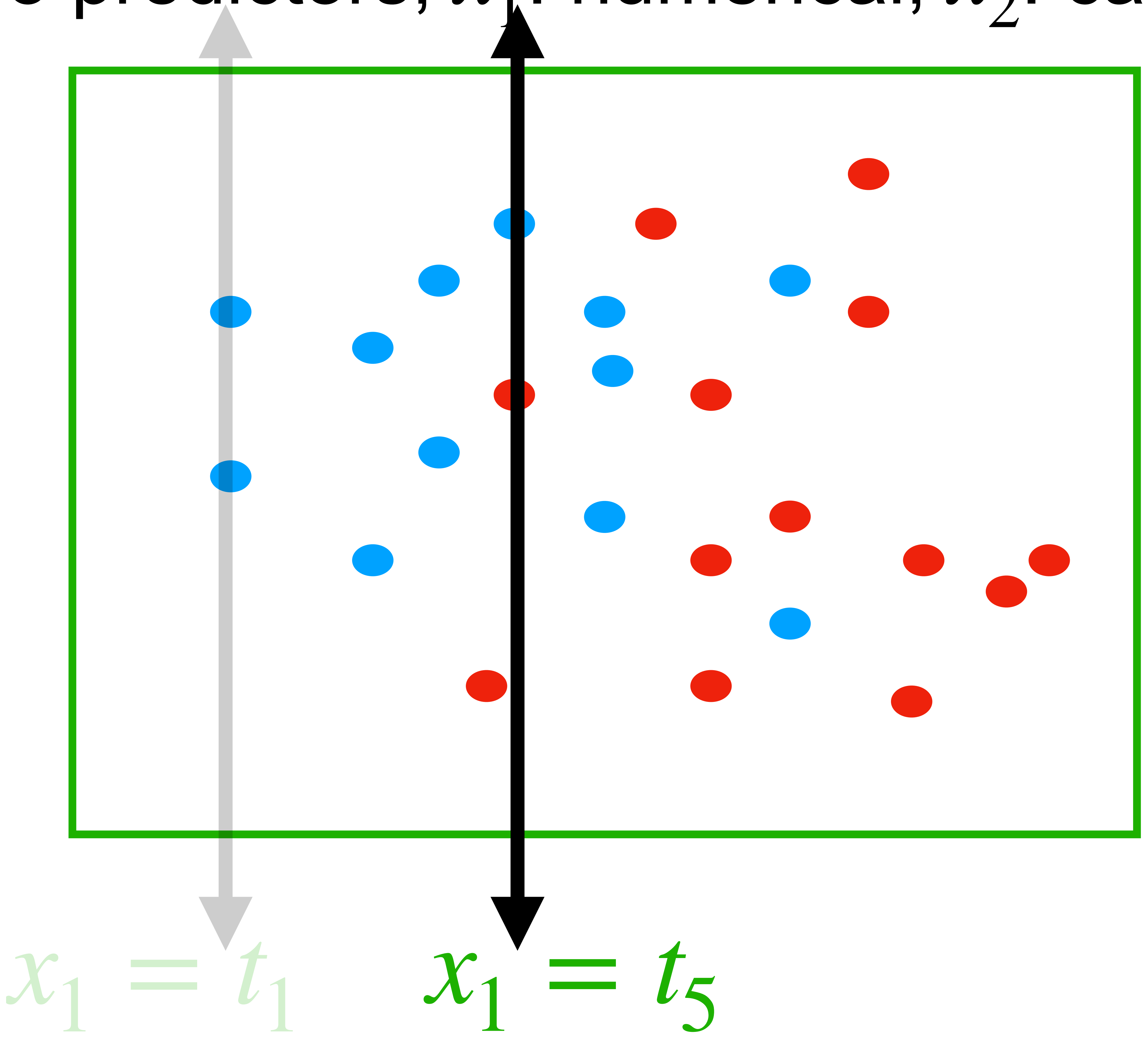

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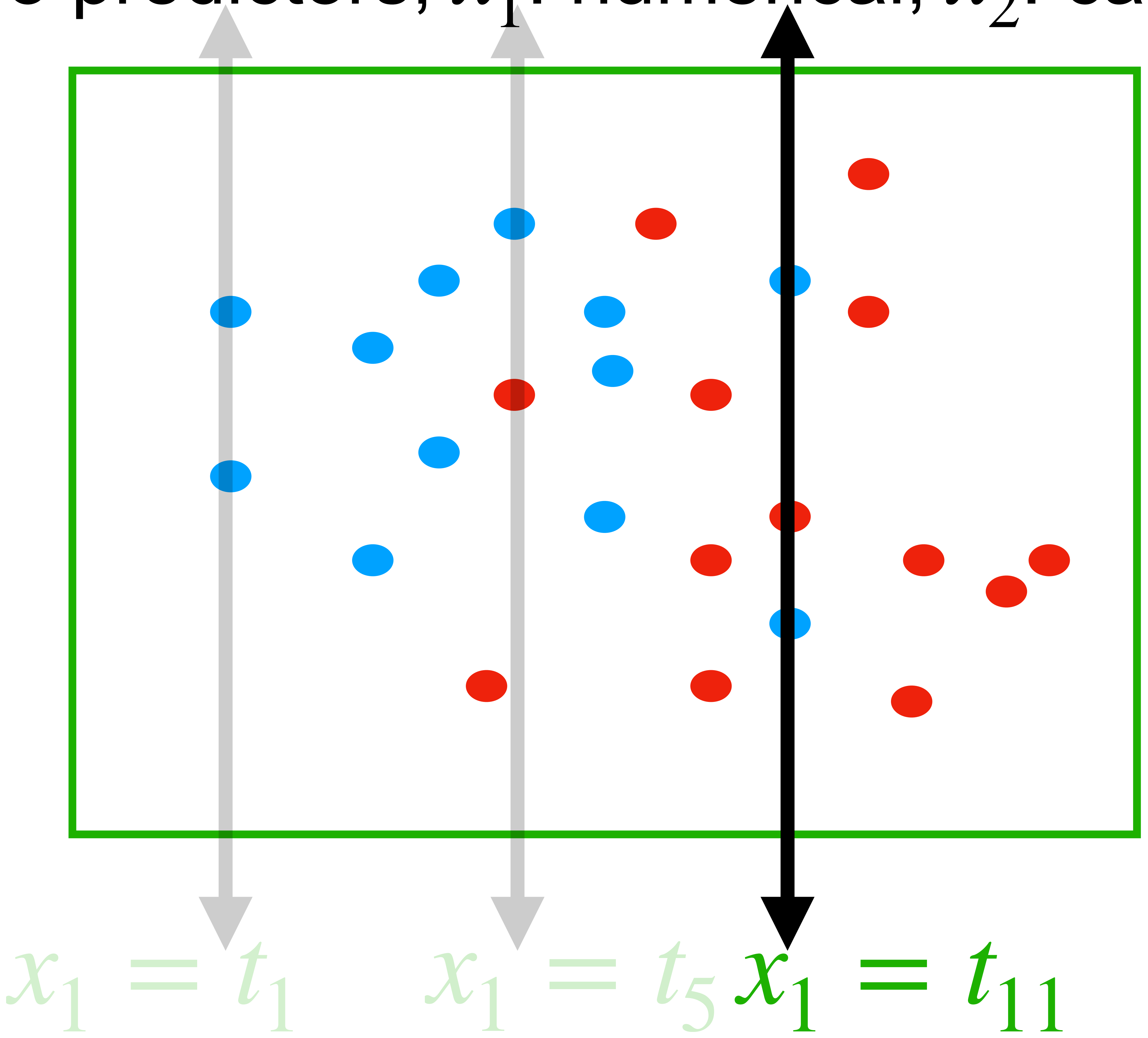

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Example thresholds:

$x_1 \leq t_1$ vs $x_1 > t_1$ OR $x_1 \leq t_5$ vs $x_1 > t_5$ OR $x_1 \leq t_{11}$ vs $x_1 > t_{11}$ …

HOW IS A DECISION TREE CONSTRUCTED?

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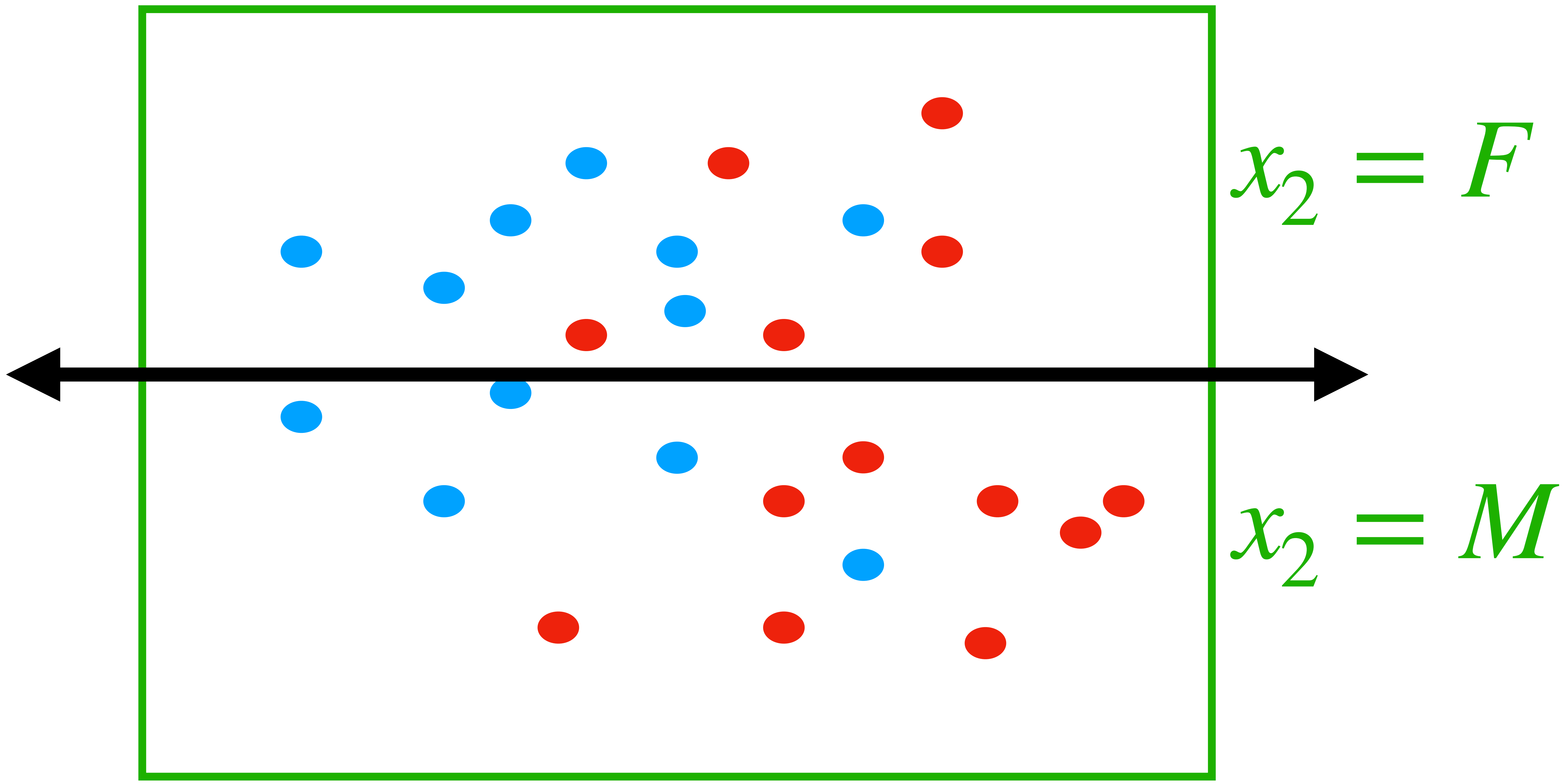

Boolean predicate at u?

Try all possible values based on predictor x_2 and record 'quality of split' for each.

HOW IS A DECISION TREE CONSTRUCTED?

How to measure quality of split?

Alternative 1:

$$Gini\ index = G = \sum_{j=1}^{K} p_j(1 - p_j) = 1 - \sum_{j=1}^{K} p_j^2$$

K : number of classes

p_j : proportion of training data points from class j.

HOW IS A DECISION TREE CONSTRUCTED?

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K : number of classes

p_j : proportion of training data points from class j.

Intuitively:

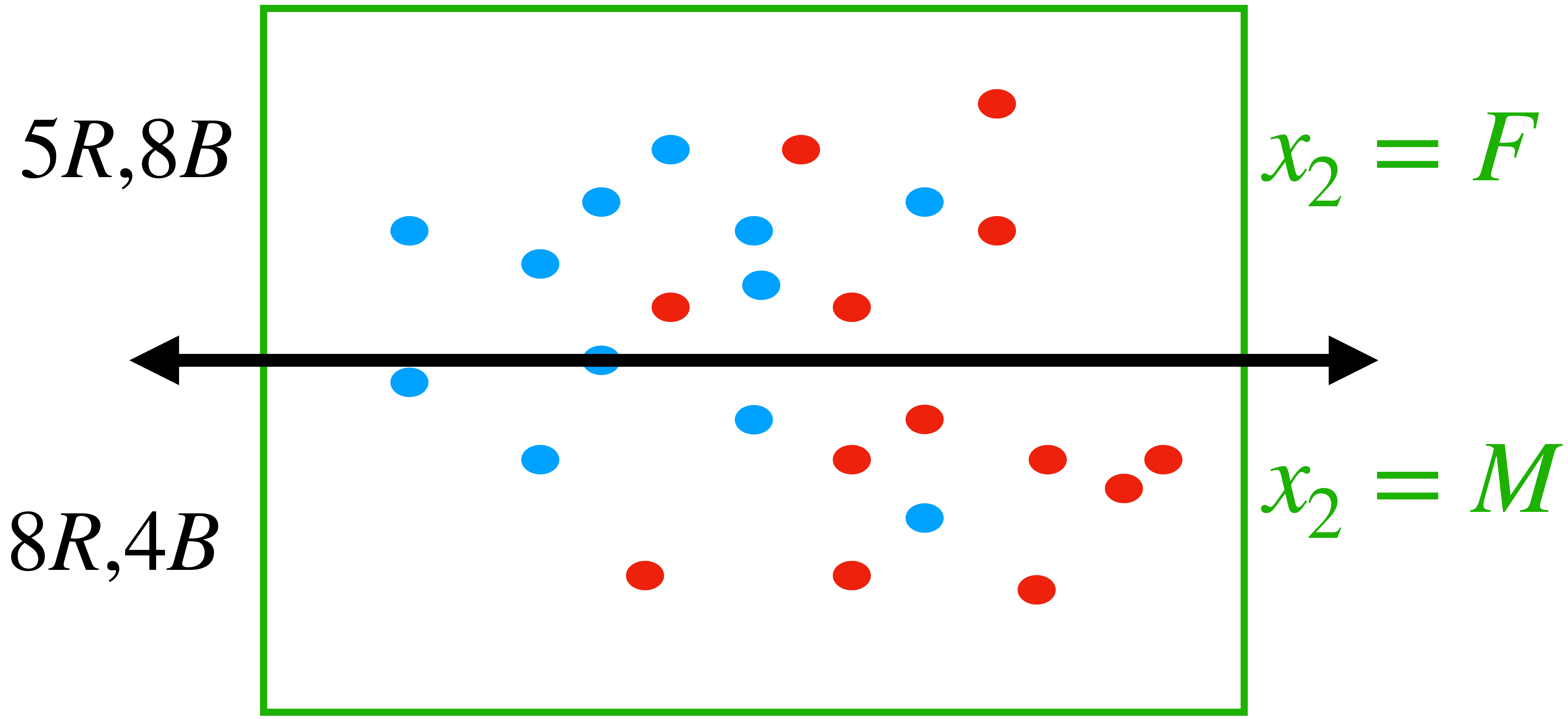
All proportions near 0 or 1 $\Rightarrow$ Small Gini index, large purity (one class dominant).

Good split: Minimizes Gini index of the split

$\underbrace{\text{Weighted-sum}}_{\text{Proportion of data points in the partition.}}$ of Gini indices of the two partitions

HOW IS A DECISION TREE CONSTRUCTED?

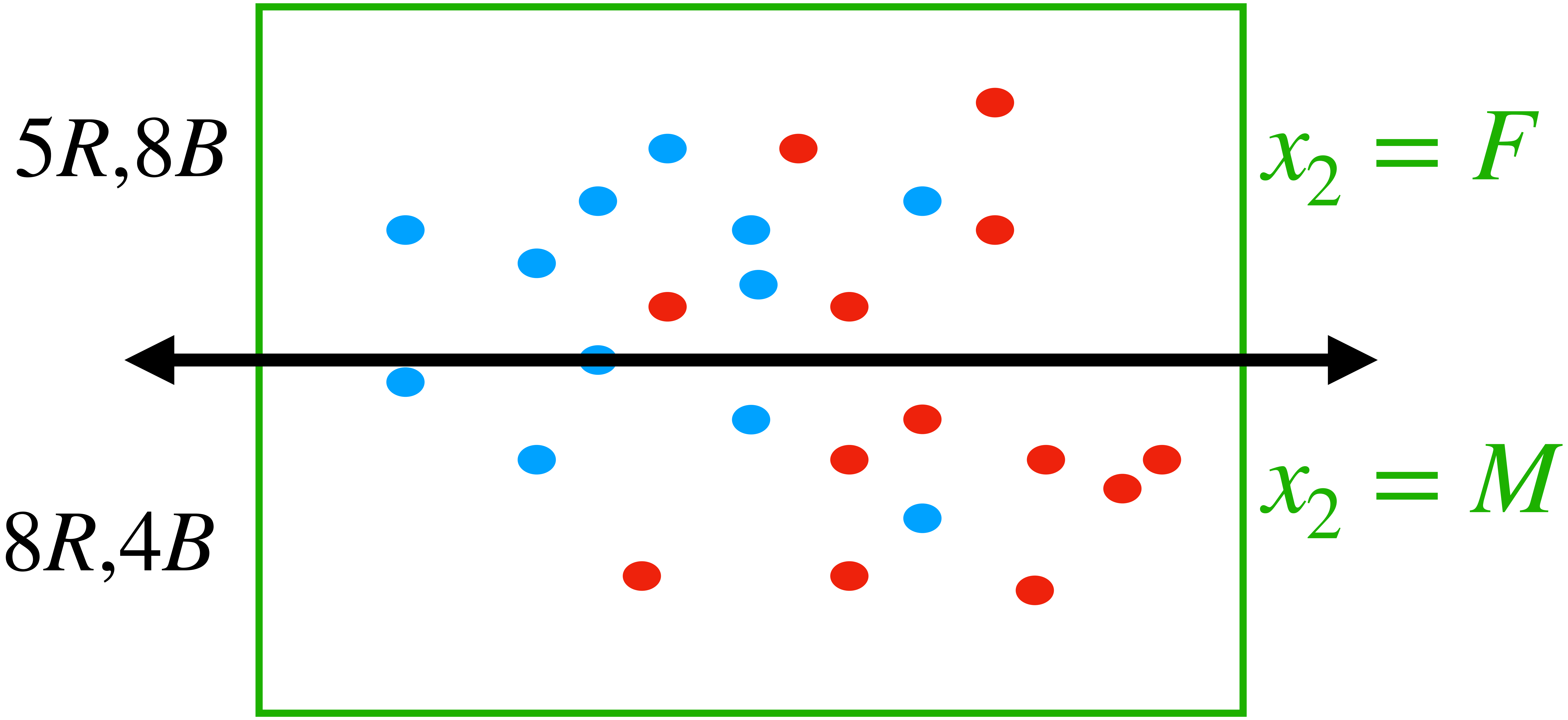
How to measure quality of split?



Example: What is the Gini index of the split?

HOW IS A DECISION TREE CONSTRUCTED?

How to measure quality of split?



Example: What is the Gini index of the split?

$$\frac{13}{25}[1-((\frac{5}{13})^2+(\frac{8}{13})^2]+\frac{12}{25}[1-((\frac{8}{12})^2+(\frac{4}{12})^2)]$$